CHAPTER FOURTEEN

Congenital Bleeding Disorders of the Vitamin K-Dependent Clotting Factors

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Contents

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Abstract

Congenital bleeding disorders of the vitamin K-dependent coagulation factors represent only about 15–20% of all congenital bleeding disorders. However, they played an important role of the history of blood coagulation.

Prothrombin was the first entity dealt with. Subsequently, in the late 1940s or early 1950s, the discovery of factor IX allowed the separation of hemophilia into two groups, A and B. In the 1950s, the discovery of factors VII and X allowed the formulation of a logic and plausible explanation for the clotting mechanism. The subsequent discovery of vitamin K-dependent proteins with an inhibitory effect on blood coagulation has further enhanced the importance of the vitamin K-dependent clotting factors.

Recently, the study of families with multiple defects of the prothrombin complex has spurred the interest in vitamin K metabolism and the γ-carboxylation system. The relevance of these studies had also an important role in the understanding the mechanism of action of other noncoagulation-related proteins. The vitamin K-dependent clotting factors represent a homeostatic mechanism at the basis of the hypercoagulability (thrombosis)—hypocoagulability (hemorrhagic) system, namely, to a mechanism that is vital for survival.

The different bleeding condition will be dealt with separately, namely, prothrombin or Factor II, Factor VII, Factor IX (hemophilia B), and Factor X deficiencies. An additional heading deals with the combined defect of the prothrombin complex, namely, combined deficiency of Factor II, Factor VII, Factor IX, and Factor X. Since, sometimes, a hemorrhagic role has been attributed to Protein Z deficiency, another vitamin K-dependent protein, this defect will also be dealt with, even though briefly.

Each deficiency has been approached in a global manner, namely, with adequate reference to history, background, prevalence, classification, hereditary pattern, biochemistry and function, molecular biology, clinical picture, updated laboratory diagnosis, prognosis, and therapy. Particular emphasis has been placed on the significance of cases with "true" deficiency [cross-reacting material (CRM negative)] and cases with abnormalities (CRM positive). The genetic, clinical, and laboratory implications of these two forms have been extensively discussed in every instance. The importance of a multiple, combined diagnostic approach that has to include whenever possible clotting, chromogenic, immunological, and molecular biology studies has been underlined. Clotting tests have to be carried out using different activating agents since results may vary, thereby indicating a different reactivity of the abnormal protein. Molecular biology techniques, alone, are unable to supply plausible diagnostic conclusions. In fact the genotype–phenotype relation has not been clarified so far for most of these bleeding conditions. Recent progress in management such as the use of recombinant factor concentrates, results of liver transplantation, and attempts at genetic therapy has been discussed. Potential complications of therapeutic measures have also been discussed.

A section dealing with future putative aims of research in this field will close the chapter.

I. Introduction

Bleeding disorders of the prothrombin complex (PC) factors, namely, Factor II, Factor VII, Factor IX, and Factor X, represent an important part of the history of blood coagulation (Greenberg and Davie, 2006). One of them, Factor IX, responsible for hemophilia B was suspected when it was shown about 60 years ago that a hemophilic plasma was corrected by the addiction of the plasma of another patient with "hemophilia" (Livingstone and Johnstone, 1963). This observation allowed to separate the hemophilia in two groups, A and B (Aggeler *et al.*, 1952; Biggs *et al.*, 1952). Prothrombin deficiency was the first coagulation defect, after hemophilia, to be known and studied as the source of thrombin.

Factor VII deficiency was described in 1951 by separating it from prothrombin deficiency (Alexander *et al.*, 1951).

Factor X deficiency was discovered in 1956–1957, and in this case too the defect was observed in one family by restudying a patient originally described as "hypoproconvertinemia" (Bachmann *et al.*, 1957; Hougie *et al.*, 1957; Kemkes-Matthes and Matthes, 2001; Telfer *et al.*, 1956).

These factors have a common origin even though a different genetic control. In fact, Factors II, VII, and X are controlled by autosomal recessive genes, whereas Factor IX deficiency (hemophilia B) has an X-linked hereditary pattern. As a consequence, both sexes are affected in cases of Factor II, Factor VII, and Factor X deficiencies, but only males are affected by hemophilia B. Exceptional description of hemophilia B in women is due to the mating of a carrier and a hemophiliac or to extreme lyonization or to rare chromosome X abnormalities (Shetty *et al.*, 2001a). The phylogenetic significance of this discrepancy is unknown, if any.

Different chromosomes are involved in the control of the synthesis of these factors.

X chromosome is involved in hemophilia B, whereas chromosome 13 controls Factor VII, Factor X, and Protein Z. Factor II depends on genes located in chromosome 11; for comparison the other two PC factors, Protein C and Protein S are under control of genes located in chromosomes 2 and 3, respectively. The significance of these different controls is not fully understood. The close location on the long arm of chromosome 13 of Factor VII, Factor X, and Protein Z genes suggests a common origin.

The PC factors have similar biochemical features; they are vitamin K-dependent glycoproteins but quite different biological half-life, very short for Factor VII, very long for Factor II, while those for Factors IX and X are in between that range. Note that the half-life of coagulant or anticoagulant factor PC has some similarities. For example, two of them, FVII and Protein C, have short half-life, whereas others, for example Protein S and Factor II, have long ones. Does this possible pairing have any significance? We do not know.

The clinical manifestations of all these deficiencies are similar and involve all structures with a prevalence of hemarthrosis for Factor IX deficiency. Brain hemorrhage is a frequent cause of death in all cases. It has been described in fact not only in Factor VII deficiency but also in all other conditions.

In general, the deficiency of these factors presents a variable gravity of bleeding manifestations that, with the exception of Factor VII deficiency, correlates fairly well with the clotting activity of the factor involved.

Factor IX and Factor VII are the least rare, whereas Factor II and Factor X are very rare coagulation disorders.

The classification of coagulation defects has become, in recent years, more difficult because of the several variants described. An important element for Factor II, Factor VII, and Factor X is the distinction between cases of "true" deficiency (concomitant decrease of activity and antigen, namely, cross-reacting material—CRM—negative cases or Type I) and abnormalities (low activity with variable, but always significantly higher, antigen levels, namely, CRM positive cases or Type II).

The classification for Factor IX is similar with the exception that some of CRM positive patients also possess a peculiar inhibitory action (hemophilia Bm) on the Tissue Factor + Factor VII system. Such inhibitory behavior has never been described for the CMRpos cases of the other clotting factors. Homozygotes for Factor II or Factor VII or Factor X deficiencies and compound or double heterozygotes for two deficiencies (hypo-hypo) have usually activity and antigen levels between 1% and 10% of normal. Factor activity above 10% of normal is usually associated with a mild spontaneous bleeding tendency, but bleeding may be severe after surgery or trauma. Heterozygotes with factor activity levels around 50% of normal are only occasionally symptomatic.

Occasional excessive bleeding especially after surgery or trauma has been described for heterozygotes for Factor II or Factor X. Heterozygotes for Factor VII deficiency are asymptomatic.

Factor IX deficiency (hemophilia B) is subdivided, as hemophilia A, in severe (Factor IX, <1%), moderate (Factor IX, 1–5%), and mild (Factor IX, 6–30%). Bleeding intensity correlates well with factor activity levels. Carriers of hemophilia B have activity level about 50% of normal and are asymptomatic.

It is puzzling that three of the four hemorrhagic factors are under autosomal genes, whereas the remaining one is an X-linked trait. This aspect, together with other major features of the PC factors, is gathered in Table 14.1.

The role played by PC in evolution is unknown.

The dependence on diet (vitamin K intake) for all these factors may have a biological significance. In subjects with poor diets, the concomitant decrease of both clotting and inhibitory factors assured a downgraded hemostatic balance.

In coumarin drugs treatment, the coagulation pattern favors hypocoagulation. However, the occasional thrombotic manifestation seen in coumarin-treated patients indicates that this hypocoagulable state sometimes fails.

We will describe the bleeding disorder of all PC factors separately. The defect that causes the concomitant decrease of all PC clotting factors will be dealt with separately, since it is a distinct clinical entity.

Combined defects of Factor II, or Factor VII, or Factor IX or Factor X with other clotting factors, will be dealt with after the description of the single deficient states.

Protein Z deficiency, even though not surely linked with a bleeding tendency, will be dealt with only briefly. Needless to say that the prothrombotic defects of the PC will not be dealt with being outside the scope of this chapter.

II. Congenital Prothrombin (Factor II) Deficiency

Congenital bleeding defects of prothrombin are one of the rarest coagulation disorders. In 1975, a preliminary search carried out by us gathered only 25 cases (Girolami *et al.*, 1975b). A more recent study in 1999 gathered about 25 cases of proven or probable hypoprothrombinemia

Table 14.1 Main features of prothrombin-complex factors

	MW	Controlling genes in chromosome	Hereditary pattern	Estimated prevalence	Approximate concentration (mg%)	Half-life (hours)	Main function	Comments
Factor II	72,000	11	AR	1:2,000,000	~10.0	~70	FXa substrate	
Factor VII	50,000	13	AR	1:500,000	~0.1	~5	TF cofactor	
Factor IX	57,000	X	XL	1:30,000	~0.4	~20	Part of prothrombinase	
Factor X	59,000	13	AR	1:1,500,000	~1.0	~30	FII activator	
Protein Z	62,000	13	AR ?	?	~0.25	~30	Unknown	Hemorrhagic effect not proven. Prothrombotic role not clear
Protein C	62,000	2	AR	1:100,000 ?	~0.4	~7	Inhibitor	
Protein S	69,000	3	AR	1:100,000 ?	~0.2	~40	Inhibitor cofactor	

Besides data for bleeding disorders, comparative data are also supplied for prothrombotic vitamin K-dependent proteins. Figures are only orientative and, often, approximate.
AR, autosomal recessive; XL, X-linked; MW, molecular weight; TF, tissue factor.

and about 40 cases of dysprothrombinemia (Girolami *et al.*, 1999a). Since that time, a few additional cases have been reported. The first reports on patients with hypoprothrombinemia go back to the studies by Quick (Quick, 1947; Quick and Hussey, 1962; Quick *et al.*, 1955).

Because of the lack of adequate antiprothrombin antiserum, the cases were thought to be cases of true deficiency, namely, hypoprothrombinemia. Subsequently, it was shown that these two patients were cases of dysprothrombinemia (Henriksen and Mann, 1988, 1989). On the contrary, the family studied by van Creveld (van Creveld, 1954) was subsequently demonstrated to be cases of true deficiency (Hart, 1965; Poort *et al.*, 1994). This restudy of old reports with a consequent different allocation to the dysforms complicates the evaluation of the literature.

In 1969, Shapiro *et al.* (1969) described the first family with dysprothrombinemia (Prothrombin Cordeza). In these patients, and in a few others subsequently reported, there was a discrepancy between clotting activity and antigen. Reviews of these variants were published in 1996 and 1999 (Collados *et al.*, 1996; Girolami *et al.*, 1999a).

The fact that even in severe cases, traces of prothrombin are found in the plasma of the patients, seems to indicate that the complete absence of prothrombin is incompatible with life. At this point, it may be relevant to note that even complete absence of antithrombin is incompatible with life.

A. Classification and prevalence

Prohrombin defects can be classified as other defects of clotting factors in:

1. True deficiency (hypoprothrombinemia) (homozygotes and heterozygotes)
2. Dysprothrombinemia (homozygotes and heterozygotes)
3. Hypo-dys or dys-dys forms (compound heterozygotes)
4. Combined or multiple defects

True prothrombin deficiency is probably the rarest coagulation disorder.

The hallmark of this defect is the concomitant decrease of both prothrombin activity and antigen. The discrepancy is usually of a few percentile figures and should be by the definition of <10%. A few of the cases listed as cases of hypoprothrombinemia have not yet been studied immunologically, so their allocation in the group of hypoprothrombinemia remains questionable. Usually no eponym is used to describe cases of hypoprothrombinemia (Table 14.2) (Baudo *et al.*, 1972; Borchgrevink *et al.*, 1959; Catanzarite *et al.*, 1997; Crossley *et al.*, 1992; De Bastos *et al.*, 1964; Degen *et al.*, 1995; Gill *et al.*, 1978; Girolami, 1971; Ikkala *et al.*, 1971; Josso *et al.*, 1962; Kattlove *et al.*, 1970; Lee and Li, 2002; Lefkowitz *et al.*, 2003; Livingstone and Johnstone, 1963; Montgomery *et al.*, 1978; Pina-Cabral and Justica,

Table 14.2 Cases of homozygous prothrombin "true" deficiency reported in the world literature

References	No. of cases	Sex	Parents consanguinity
Hart *et al.* (1965), Poort *et al.* (1994), and van Creveld (1954)	6	4 M; 2 F	No
Borchgrevinck *et al.* (1959)	1	F	Yes
Josso *et al.* (1962)	1	M	Yes
Pool *et al.* (1962)	1	M	No
Livingstone and Johnstone (1963)	1	M	?
De Bastos *et al.* (1964)	3	2 M; 1 F	Yes
Girolami *et al.* (1971)	1	F	Yes
Kattlove *et al.* (1970)	2	1 M; 1 F	?
Girolami *et al.* (1970)	1	M	Yes
Girolami *et al.* (1970)	1	M	No
Ikkala *et al.* (1971)	1	F	No
Baudo *et al.* (1972)	1	M	Yes
Piňa-Cabral *et al.* (1973)	1	M	No
Gill *et al.* (1978) and Tamary *et al.* (1997)	1	F	?
Montgomery *et al.* (1978)	1	M	?
Catanzarite *et al.* (1997)	1	F	No
Poort *et al.* (1997)	1	F	No
Sun *et al.* (1999)[a]	1	F	Yes
Lee and Li (2002)	1	M	?
Lefkowitz *et al.* (2003)	4	3 M, 1 F	?
Wang *et al.* (2004)	1	F	Yes

The list may be incomplete.
[a] Patient has same mutation as that previously reported by Poort *et al.* (1994), namely, Tyr 44Cys.
F, female; M, male.

1973; Pool *et al.*, 1962; Poon, 1981; Poort *et al.*, 1994, 1997; Rubio *et al.*, 1983; Strijks *et al.*, 1999; Valls-de-Ruiz *et al.*, 1987).

Dysprothrombinemia are usually indicated according to the city or area where they were first described. A few are unnamed or a tentative definition has been attributed by us for the sake of completion (Akhavan *et al.*, 2003; Bezeaud *et al.*, 1984, 1979; Board *et al.*, 1982; Board and Shaw, 1983; Dumont *et al.*, 1987; Friezner Degen *et al.*, 1995; Girolami *et al.*, 1974a, 1977, 1978; Henriksen and Mann, 1988; Henriksen *et al.*, 1998; Huisse *et al.*, 1985; Iwahana *et al.*, 1992; James *et al.*, 1994, 1995; Josso *et al.*, 1971, 1982; Kahn and Govaerts, 1974; Lefkowitz *et al.*, 2003; Miyata *et al.*, 1987, 1995; Montgomery *et al.*, 1980; Morishita *et al.*, 1992; O'Marcaigh *et al.*, 1996; Owen *et al.*, 1978; Poon *et al.*, 1979, 1981; Rocha *et al.*, 1986; Ruiz-Saez *et al.*, 1986; Sabharwal *et al.*, 1992; Shapiro *et al.*, 1973; Smith *et al.*, 1981; Sun *et al.*, 2001; Tamary *et al.*, 1997; Weinger *et al.*, 1980).

The hallmark of these conditions is a discrepancy between clotting activity and antigen level. The latter is always higher than the clotting counterpart and sometimes perfectly normal (Akhavan *et al.*, 2003).

Several families with a sufficiently proven prothrombin abnormality have been described so far. As a matter of fact, dysprothrombinemias seem more frequent than true hypoprothrombinemia do. Most of these conditions appear to be distinct entities. However, some appear to be identical to conditions previously described under different eponyms. For example, prothrombin Dhahran seems identical to prothrombin Padua, and therefore they should be dealt with together. The same is true for prothrombin Tokushima and prothrombin Molise. Furthermore, prothrombin Frankfurt seems identical to prothrombin Salakta. Finally, prothrombins Madrid and Obihiro seem identical to prothrombin Barcelona. Since prothrombins Padua, Molise, Salakta, and Barcelona were originally described before Dhahran, Tokushima, Frankfurt, Madrid, or Obihiro, respectively, the first eponym will be used to describe these abnormalities, unless different mutations are demonstrated.

Prothrombin Mexico City could be excluded from the dysprothrombinemia group and be included among the hypoprothrombinemias, because of the concomitantly low levels (about 10%) of the antigen and activity observed in this deficiency. The same is true for prothrombin Philadelphia which is characterized by 8% activity and 4% antigen level. This picture, by definition, is not compatible with the basic features of an abnormality (Table 14.3).

Prothrombin abnormalities may be, tentatively, classified according to the site of the defects as follows: defect in the activation mechanism or defects in the thrombin molecule. As is discussed later, the latter group can be subdivided into two subtypes, namely, a first subtype in which there is an abnormal interaction of thrombin with fibrinogen and a second one with a defective catalytic activity.

Molecular biology analysis has allowed further classification of the defect, even though an exact genotype–phenotype relation is not completely clear yet.

B. Biochemistry

Human prothrombin is a glycoprotein with a molecular weight of about 70,000 Da, and with a known amino acid composition. It is synthesized at the hepatocyte level in the presence of vitamin K. The plasma concentration is about 10 mg%, and the biological half-life is about 70 h (Friezner Degen, 1995; Shapiro and McCord, 1978). The circulating mature protein is a

Table 14.3 Cases of prothrombin abnormalities described in the world literature

Refererences	Eponym	Type of defect	Sex	Parents consanguinity
Quick *et al.* (1947, 1955), Quick and Hussey (1962), Owen *et al.* (1978), and Henriksen and Mann (1988, 1989)	Quick I, II	Comp. Het.	1 M; 1 F	?
Shapiro *et al.* (1969)	Cardeza	Het.		?
Josso *et al.* (1971)	Barcelona	Hom. and Het.	4 M; 2 F	No
Girolami *et al.* (1974)	Padua	Het.	4 M; 1 F	No
Kahn and Govaerts (1974)	Brussels	Hom. and Het.	4 M; 1 F	Yes
Shapiro *et al.* (1973)	San Juan	Comp. Het.	5 M; 2 F	?
Girolami *et al.* (1978b)	Molise	Comp. Het.	1 M; 1 F	Yes
Poon *et al.* (1979, 1981)	Birminghan I and II	?	?	?
Montgomery *et al.* (1980) and Lefkowitz *et al.* (2003)	Denver	Hom. and Het.	2 M; 1 F	No
Weinger *et al.* (1980)	Houston	?	1 M	?
Smith *et al.* (1981)	Gainesville	Het.	2 F	?
Josso *et al.* (1982)	Metz	Comp. Het.	3 M; 9 F	?
Rubio *et al.* (1983)	Habana	Comp. Het.	3 M; 1 F	?
Board and Shaw (1983)	Canberra	?	?	?
Bezeaud *et al.* (1984)	Salakta	Hom.?	1 M; 1 F	?

(*continued*)

Table 14.3 (*continued*)

Refererences	Eponym	Type of defect	Sex	Parents consanguinity
Huisse *et al.* (1985)	Clamart	Het.	2 F	No
Rocha *et al.* (1986) and Akhaven *et al.* (1999)	Segovia	Hom. and Het.	2 M; 1 F	Yes
Ruiz-Saez *et al.* (1986) and Sekine *et al.* (2002)	Perija	Hom. and Het.	2 M; 2 F	Yes
Dumont *et al.* (1987)	Poissy	Hom. and Het.	2 M; 2 F	Yes
Lutze *et al.* (1989)	Magdeburg	Het.	4 M; 3 F	?
Morishita *et al.* (1992)	Himi	Comp. Het.	1 M; 3 F	No
Strijks *et al.* (1999)	Nejmegen	Hom. and Het.	1 M	Yes
Akhavan *et al.* (2000)[a]	G	Hom.	1 M	No
	I	Hom.	1 F	Yes
Akhavan *et al.* (2000)[a]	A	Hom.	1 F	Yes
	B, C1, C2	Comp. Het.	2 F, 1 M	Yes
	D	Hom.	1 F;	No
	E, F	Hom.	1 F, 1 M	Yes, No
	H	Hom.	1 F	No
	J1, J2, K	Hom.	1 F, 2 M	No
Lefkowitz *et al.* (2003)	Puerto Rico 1	Hom.	1 M; 1 F	Yes
	Puerto Rico 2	Hom.	n.r.	Yes

The list may be incomplete.
Prothrombins Corpus Christi, Madrid and Obihiro, Daharan, Tokushima, and Frankfurt are not listed, since they seem identical to prothrombins Quick I, Barcelona, Padua, Molise, and Salakta, respectively. Prothrombins Mexico City and Philadelphia have not been listed because they appear cases of true deficiency (concomitant decrease of activity and antigen). Het., heterozygotes; Hom., homozygotes; Comp. Het., compound heterozygotes; M, male; F, female; n.r. not reported.
[a] Kinders indicated with letters as reported by authors.

single chain glycoprotein of 579 residues (Friezner Degen, 1995; Lanchantin *et al.*, 1968; Shapiro and McCord, 1978).

The prothrombin molecule contains five domains: the propeptide (residues −43 to −1), the Gla domain (residues 1–40), a kringle domain (residues 41–155), a kringle 2 domain (residues 156–271), and a serine protease domain precursor (residues 272–579). Prothrombin is activated by Factor Xa on the surface of activated platelets in the presence of Factor V. Fragment 1–2, a large activation peptide, is thus obtained as the result of the cleavage of the peptides Ag 271–Th 272. The remainder of the molecule is not yet enzymatically active and is termed Prothrombin 2. The second cleavage splits the prothrombin molecule at residue 320, forming a light and a heavy chain of thrombin without releasing them as separate entities because of a single disulfide bond. This two-chain molecule is α-thrombin, the active enzyme.

Other intermediate products are formed during the conversion of prothrombin into thrombin. The type and the relative presence of these intermediate products vary with the type of activation. The analysis of such activation pattern is beyond the scope of this chapter.

From a biochemical standpoint, congenital dysprothrombinemias could be classified into two groups, namely, defects in the activation mechanism; and defects in the thrombin molecule. The first group includes prothrombins San Juan (inability to bind Ca^{2+}), Barcelona (defective cleavage by FXa at site Arg 271–Thr), and Clamart (defective cleavage by FXa at Arg 321–Ile). In these cases and other similar dysprothrombinemias, prothrombin activation is abnormal and slow but thrombin, once formed, appears to be normal.

In the second group, the defect resides not in the activation mechanism but in the final product, namely, thrombin. As far as this type is concerned, two subtypes are usually recognized, one in which the defect lies in the interaction of thrombin with fibrinogen and the other in which there is a defect in the catalytic region of thrombin. Examples of the first subtype (defective interaction with fibrinogen) are: prothrombins Quick, Salakta, and Himi. Examples of abnormality subtype 2 (defective catalytic region) are prothrombins Molise (Tokushima), Habana, and Metz (Table 14.4).

C. Genetics and hereditary transmission

The gene encoding prothrombin is located on chromosome 11 (p11-p12). The gene contains 14 exons and 13 introns (Degen and Davie, 1987). The introns serve to separate the domains of the protein. Exons 1 and 2 encode the propeptide; the Gla domain is encoded by Exon 2 and is composed of the 10 glutamic acid residues that are carboxylated by vitamin K-dependent enzymes. These residues play a fundamental role in Ca-binding on cellular membranes.

Kringle 1 and 2 domains are encoded by Exons 3–6 and 7–8, respectively. The function of these domains is still unclear, even though it seems that kringle 2 is involved in the binding of Factor Xa.

Table 14.4 Prothrombin activity and antigen in the inherited dysprothrombinemias reported in the world literature

Abnormality	Activity (% of normal)	Antigen (% of normal)	Nature of defect	Bleeding
Cardeza	50	100	Prethrombin 2 region	No
Barcelona (Madrid, Obihiro)	12 (3,18)	100 (100, 105)	Factor Xa cleavage	Yes
Padua (Dhahran)	50 (6)	100 (100)	Factor Xa cleavage	Yes
Brussels	46	88	?	Yes
San Juan	20	93	Calcium-binding site	Yes
Molise (Tokushima)	11 (11)	45 (42)	Thrombin region–reduced interaction with fibrinogen	Yes
Quick I and II	<2	34	Thrombin region	Yes
Birmingham I, II	8–10 (<3)	24–30 (8–14)	?	Yes
Denver	<1 (5)	13 (21)	?	Yes
Houston	5–9	51	Factor X cleavage	Yes
Gainesville	23	70	Factor Xa cleavage	Yes
Metz	10	50	Thrombin region	Yes
Canberra	?	?	Pro-region	No
Salakta (Frankfurt)	16 (13)	100 (91)	Thrombin region	No (Yes)
Segovia	7–20	100	Fragment 2 defect?	Yes
Perija	2	70	Factor Xa cleavage	Yes
Clamart	50	100	Factor Xa cleavage	No
Poissy	2	47	?	Yes
Habana	10	50	?	Yes
Magdeburg	44	90	?	Yes
Himi I and II	10	88	Thrombin region	Yes
Kindred G	5–23	71	Thrombin region	Yes
Kindred I	1	71	Thrombin region	Yes
Puerto Rico, Kindred 3	9	35	Activation	Yes
Kindreds 4 and 5	15–19	28–40	Activation	Yes

The list may be incomplete.
Prothrombin Corpus Christi is identical to prothrombin Quick I.
Numbers or letters indicate kindreds as reported by authors. Levels are approximated to the closest number.

The thrombin-α chain, which is formed after FXa cleavage at 271–272 and 320–321 residues, is encoded by Exons 8 and 9 (catalytic area), while the β chain is encoded by Exons 9–14.

The overall evaluation of all mutations described in prothrombin deficiency and abnormalities does not allow yet a satisfactory comprehension of genotype–phenotype and of structure–function relations.

Prothrombin deficiency is transmitted as an autosomal recessive trait. The same is true for dysprothrombinemias. Compound defects are characterized by an association of a heterozygous "true" deficiency and a heterozygous abnormality, or by two, distinct abnormalities. These conditions are known as compound heterozygosis.

It is interesting to note a clear prevalence of patients with a Latin background. About 70% of all patients with prothrombin defects have come from countries of Latin background (Cardeza, Barcelona, Padua, Segovia, San Juan, Habana, Puerto Rico, etc.). The significance of this observation remains to be clarified (Fig. 14.1).

Recently, the genetic analysis of cases of true hypoprothrombinemia has been carried out. The large family originally described by van Creveld was shown to have a missense mutation from Tys 44 to Cys. The reason why this point mutation is associated with very low levels of both prothrombin activity and antigen remains to be clarified. At present, several other cases of hypoprothrombinemia were investigated by gene analysis as is shown in Table 14.5. Mutations involving several exons have been described. Most frequently they involve the catalytic area (Exons 8–14). A clear genotype–phenotype relation has not always been consistent. All these studies indicate the great heterogeneity of human prothrombin defects. In fact, mutation in different exons may cause similar CRMneg forms.

Several dysprothrombinemias have also been investigated from a molecular point of view. The different mutations seen in these dysprothrombinemias are gathered in Table 14.6. Occasionally, the same mutations have been found in prothrombin, originally labeled with different eponyms. It is clear that, until it is proven otherwise, the abnormalities have to be considered identical. Again, it is evident that mutation in different distant exons may also give origin to CRMpos forms (Molise and prothrombin Tokushima).

D. Clinical picture

True deficiency of the homozygote levels is always characterized by several and sometimes by severe bleeding manifestations. Intracerebral bleeding and hematomas have been reported. Table shows the bleeding symptoms presented by the patients reported in the literature. Cerebrovascular bleeding seems particularly life threatening.

Altogether, there is no possibility to reach a diagnosis of prothrombin defects on clinical grounds; the bleeding manifestations are varied, none is

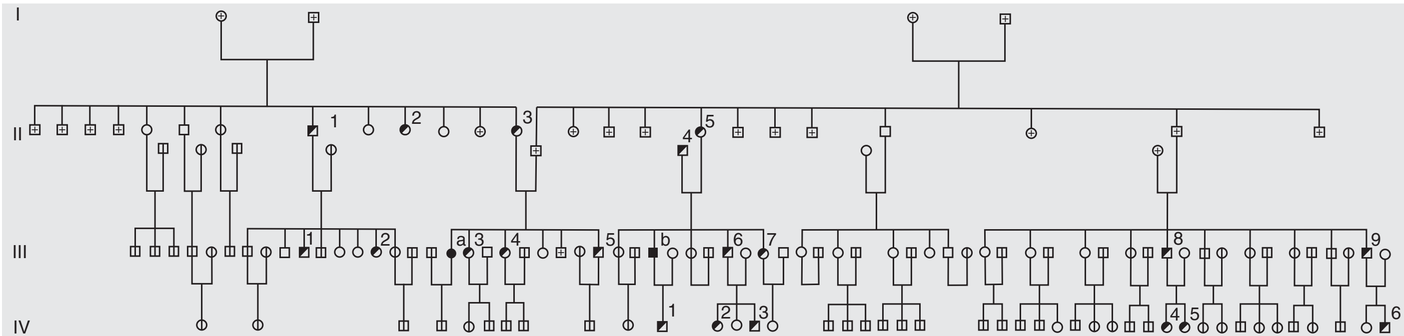

Figure 14.1 Large kindred with congenital hypoprothrombinemia (''true'' prothrombin deficiency). The hereditary transmission is autosomal recessive. The same is true for congenital dysprothrombinemias. Full-darkened symbols = homozygotes (IIIa, female and IIIb male); half-darkened symbols = heterozygotes; ○ or □ died, ○ or □ alive, not studied. The two homozygotes were first cousins. The son of homozygote IIIa was subsequently found, as expected, to be heterozygote.

Table 14.5 Molecular defects seen in cases of true deficiency so far studied

References	Genotype	Mutation (exon)
Tamary *et al.* (1997)	Comp. Het. (hypo/hypo)	Del 249/250 + Arg34Trp (8, 10)
Poort *et al.* (1997)	Comp. Het. (hypo/hypo)	Cys138Tyr + Trp357Cys (6, 10)
Poort *et al.* (1994) and Sun *et al.* (1999)	Hom.	Tys-44 Cys (3)
Akhavan *et al.* (2000)		
Case A	Hom.	Arg -1 Gln (2)
Case B, C1, C2	Hom.	Arg -2 Trp (2)
Case D	Comp. Hom.	Asp118Tyr; Arg220Cys (6, 7)
Case E, F	Hom.	Del Lys 301/302 (9)
Case H	Hom.	Arg538Lys (14)
Case J1, J2-K	Hom.	Ser394Arg (10)
Lefkowitz *et al.* (2003)		
Kindred 1	Comp. Het. (hypo/hypo)	Arg457Gln + exon 5/6 splice junction
Kindred 2	Comp. Het. (hypo/hypo)	Arg457Glu deletion + 5 base pair deletion (13)
Wang *et al.* (2004)	Hom.	Glu 29 Gly (2)

The list may be incomplete.
Eponyms occasionally attributed to cases of "true" hypoprothrombinemia have been omitted. It is evident that mutations in different exons may cause similar CRM negative forms.
Letters indicate kindreds as reported by Authors. CRM, cross-reacting material; Het., heterozygotes; Hom., homozygotes; Comp. Het., compound heterozygotes.

typical and the picture may be similar to that seen in other clotting factor deficiencies (Table 14.7).

Heterozygotes are usually asymptomatic but, occasional excessive bleeding after tooth extractions, tonsillectomy or after surgical procedures have been observed (Table 14.7). In this regard, it has to be remembered that patients heterozygous for Factor VII deficiency are instead always asymptomatic.

Bleeding tendency in dysprothrombinemias is much more variable and usually less severe than in true deficiency. Altogether, there is a fair correlation between prothrombin activity and the clinical picture.

At least two prothrombin abnormalities (prothrombin Cardeza and prothrombin Salakta–Frankfurt) do not seem to be associated with bleeding.

Table 14.6 Molecular defects seen in congenital dysprothrombinemias so far investigated

Variant	Genotype	Molecular defect (exon)
Barcelona (Madrid/ Obihiro)	Hom. and Het.	Arg271Cys (8)
Padua (Dhahran)	Hom. and Het.	Arg271His (8)
Molise (Tokushima)	Comp. Het. (hypo-dys)	Arg418Trp + stop codon 174 (11, 7)
Quick I (Corpus Christi)	Comp. Het. (dys-dys)	Arg382Cys (9)
Quick II	Comp. Het. (dys-dys)	Cly558Val (12)
Salakta (Frankfurt)	Hom. and Het.	Glu acid 446Ala (12)
Himi I	Comp. Het. (dys-dys)	Met337Thr (9?)
Himi II	Comp. Het. (dys-dys)	Arg388His (9?)
Prothrombin 3 Camberra	Het.	Glu acid 157Lys (3)
Bangladesh	Comp. Het. (hypo-dys)	Arg271His + Hip518Arg (8, 14)
Nijmegen	Hom. and Het.	Glu7Lys (2)
Greenville	Het.	Arg517Glu (13)
Segovia	Hom. and Het.	Glu319Arg (9)
Perija	Hom. and Het.	Glu548Ala (14)
San Antonio	Het.	Arg320His (9)
Denver	Comp. Het. (hypo-dys)	Glu300Lys + Glu309Lys (9)
Scranton	Het.	Lys556Thr (14)
No eponym (Akhawan *et al.*)		
G	Hom.	Gly 330 Ser (9)
I	Hom.	Arg 382 His (10)
Puerto Rico 1	Comp. Het.	Arg457Gly + Glu16Gly (12;2)
Puerto Rico 2	Hom.	Arg457Glu (12)

The list may be incomplete. In this case too, the mutations claimed to be responsible for the CRM positive forms, involve several exons.
Letters indicate kindred as reported by authors. CRM, cross-reacting material; Het., heterozygotes; Hom., homozygotes; Comp. Het., compound heterozygotes.

Table 14.7 Hemorrhagic manifestations observed in 26 patients with congenital hypoprothrombinemia as reported in the literature

Hemorrhagic manifestation	No. of patients	No. of positive patients	%
Umbilical cord hemorrhages	26	4	15.4
Epistaxis	26	14	53.8
Hematomas and ecchymosis	26	17	65.3
Gingival bleeding	26	3	11.5
Bleeding after tooth extraction	26	9	34.6
Gastrointestinal or abdominal hemorrhages	26	3	11.5
Hemarthroses	26	11	42.3
Hematuria	26	2	7.6
Menorrhagias, metrorrhagias	3[a]	3	100.0
Post-partum hemorrhages	2	2	100.0
Cerebral or intracranial hemorrhages	26	3	11.5

[a] Post-puberal women.

E. Diagnosis

A correct diagnosis of prothrombin deficiency can be reached only by laboratory investigation.

Routine laboratory tests such a PT and PTT are usually slightly or variablly prolonged. Note that, contrary to what is observed for defect of Factors V, VII, or X, these tests may be normal or only borderline in isolated prothrombin deficiency. This may explain, at least in part, the rarity of the defect, since many cases have probably gone undetected because routine tests were so minimally altered that no further study was carried out.

Prothombin-time-derived tests, such as Thrombotest, Normotest, or PP test, are usually more prolonged. Once a suspicion is raised, a specific prothrombin assay has to be carried out. Several tests are available (Denson *et al.*, 1971; Hjort *et al.*, 1955). The one-stage standard assay using tissue thromboplastin as activating agent and a prothrombin-free substrate is the most widely used. Provided the substrate is adequate, the test is reliable and sufficient for screening purposes (Girolami *et al.*, 1974b). The two-stage method is rarely used at present (Ware and Seegers, 1949).

Homozygous patients with "true" deficiency have usually prothrombin activity and antigen levels of less than or about 10% of normal, regardless of the method used (Table 14.8). Complete absence of prothrombin seems incompatible with life. Patients heterozygous for the same defect usually have prothrombin levels between 40% and 60% of normal. On repeated assays, a good separation of heterozygotes from normal, unaffected members

Table 14.8 Prothrombin level (%) in three homozygous patients with true hypoprothrombinemia seen in Padua as determined by different techniques

Method	Patient 1	Patient 2	Patient 3
Classical one-stage method (Quick, 1946)	15	9	16
Iowa two-stage method (Ware and Seegers, 1949)	13	11	12
Notechis scutatus scutatus viper venom (Jobin and Esnouf, 1966)	14	12	16
Taipan viper venom (Denson *et al.*, 1971)	12	10	11
Echis carinatus venom (Girolami *et al.*, 1976)	10	11	12
Chromogenic (S2238)	13.6	14.6	9
Staphylocoagulase (Soulier and Prou-Wartelle, 1966)	9	10.5	11
Immunological (Mancini *et al.*, 1965)	12.5	11.5	11.5

of the family may be achieved. In a group of 26 heterozygotes, we found an average of 51% (range 43–75). These values were easily distinguishable from a group of normal relatives (mean 98; range 84–130).

Other tests, using other activating substances, are needed to screen for potential abnormalities. Regardless of the assay used, the activity level in case of "true" deficiencies should be about the same. Should one encounter a discrepancy in the activity level, a dysprothrombinemia should be suspected. In this regard, the Echis Carinatus venom assay may be important, since it has been reported to be normal in a few dysprothrombinemias (Girolami *et al.*, 1984). It has to be remembered that prothrombin-activating agents used in clotting tests have different methods of converting prothrombin into thrombin. This is particularly true for viper venoms. The four major venoms used in prothrombin assays have, in fact, different requirements. Tiger snake venom behaves as a tissue thromboplastin requiring phospholipids, Factor V, and calcium. On the other hand, Echis Carinatus venom converts prothrombin directly without depending on other factors. An intermediate position is shown by Taipan viper venom and Australian brown viper venom which do not require Factor V, but do require phospholipids and calcium for optimal activity (Table 14.9) (Denson *et al.*, 1971; Masci *et al.*, 1988).

Chromogenic substrates have been used for prothrombin assay (Girolami *et al.*, 1980b). In this regard, it has to be kept in mind that clotting activity is not always equivalent to chromogenic activity. In fact, chromogenic prothrombin assay has been normal in prothrombin Padua and higher than expected in prothrombin Molise.

Table 14.9 The variable requirements of several one-stage Factor II assays

Assay	Factor V	Phospholipid	Ca^{2+}	Activating agents
Classical one stage	+	+	+	Tissue thromboplastin
Tiger snake venom	+	+	+	Tiger snake venom
Taipan viper venom	No	+	+	Taipan viper venom
Textarin time	No	+	+	Pseudonaja textiles venom
Echis Carinatus venom	No	No	No	Echis Canaritus venom
Staphylocoagulase	No	No	No	Staphylocoagulase

An immunological investigation is essential (electroimmunoassay, ELISA, radial-immunodiffusion, laser nephelometer, etc.) (Girolami *et al.*, 1982b; Mancini *et al.*, 1965). Without a careful antigen evaluation, patients with prothrombin deficiency cannot be adequately classified. Even molecular biology studies are unable to substitute for an antigens assay, since no specific group of mutations has been associated with either surely CRMpos or CRMneg forms.

The staphylocoagulase is a clotting method that employs as activating agent staphylocoagulase, derived from certain strains of *Staphylococcus aureus* cultures. It complexes with prothrombin to form thrombin coagulase that, in the presence of an excess of fibrinogen, yields a clotting time that depends only on the level of prothrombin antigen (Soulier and Prou-Wartelle, 1966).

Difference between clotting and antigen activity <15% is probably of little significance, if any. One may still be dealing with a hypoprothrombinemia. Discrepancies higher than this support dysprothrombinemias.

If a dysprothrombinemia is suspected, further studies are indicated. Prothrombin clotting assay using different activation agents (Taipan viper venom, Echis carinatus venom, etc.) can be used. Bidimensional immunoelectrophoresis on plasma and serum to detect abnormal migration of whole prothrombin or prothrombin split products in serum is another alternative. Some dysprothrombinemias will show abnormal migration of plasma prothrombin, either less anodic or more anodic (Table 14.10). Other dysprothrombinemias show an abnormal pattern in the migration of serum prothrombin. Molecular biology techniques are absolutely indicated to further characterized the defect (Fig. 14.2). These have become almost routine when investigating new cases. Database of mutations is being continuously enriched. Unfortunately, the genotype–phenotype relation

Table 14.10 Plasma (or serum) prothrombin immunoelectrophoretic patterns of some abnormal prothrombins

Normal	Less anodic (slow)	More anodic (fast)
Padua (serum abn.)	Prothrombin 3 (Canberra)	Barcelona
Molise (serum abn.)	Segovia	San Juan (serum abn.)
Cardeza (serum abn.)	Denver	Habana (serum abn.)
Clamart	–	–
Houston	–	–

Abn. = abnormal.

is not completely clear yet (Akhavan *et al.*, 2003). Recently, three-dimensional putative models have been constructed (Akhavan *et al.*, 2000).

F. Prognosis

Prognosis is guarded for homozygotes and good for heterozygotes. Because of the frequency of cerebrovascular bleeding in homozygote patients, morbidity and mortality are high. The eldest alive patient is a 61-year-old female (Girolami, 1971). In the case of dysprothrombinemias, prognosis seems better. In fact, the eldest patient described is a 75-year-old man (Weinger *et al.*, 1980). There are few studies available concerning the effect of a prophylactic approach in these patients. The long survival time of infused prothrombin could justify this approach (Biggs and Denson, 1963).

G. Therapy

Substitution therapy for prothrombin deficiency is carried out by administering plasma, prothrombin concentrates, or both. Only homozygotes need prompt treatment in case of bleeding or adequate prophylaxis before surgical procedures. Since no isolated Factor II concentrate exists, concentrates of PC factors are used (Lechler, 1999). The problem with these concentrates results from the fact that they are often titrated in Factor IX content and not in Factor II. This may cause some discrepancy between the amount (units of Factor IX) administered and the *in vivo* recovery (Factor II level). More recent Factor IX concentrates, monoclonal or recombinant, should be used since these allegedly do not contain any Factor II. The long half-life of infused prothrombin (about 70 h) allows infusion every 2–3 days with a satisfactory yield (Biggs and Denson, 1963; Soulier *et al.*, 1962). Oral or intravenous antifibrinolytic agents have also been used. Because of the rarity of the condition, there are no reports on liver transplants or gene therapy attempts. The oral contraceptives have been shown to be beneficial with a sharp decrease of menometrorrhagia and improved hematocrit.

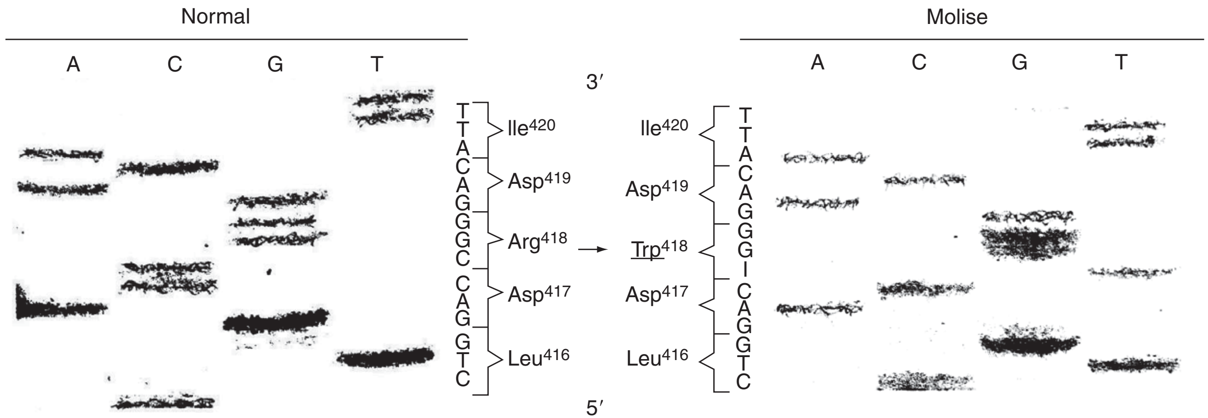

Figure 14.2 Autoradiographs showing the prothrombin Molise gene defect. A missense mutation, C to T at nucleotide position 9,490 in Exon 11, results in coding for Trp (TGG) instead of Arg (CGG) at amino acid position 418.

H. Combined defects

The most important combined defect of the prothrombin is the association with Factors VII, IX, X, and Proteins C and S due to the malfunction or absence of one of the enzymes involved in vitamin K metabolism. This combined defect will be dealt with in a separate heading. Casual associations of prothrombin with other isolated clotting factors have never been described.

III. Congenital Factor VII Deficiency

The first case of congenital deficiency of Factor VII (FVII) was described by Alexander (Alexander *et al.*, 1951), and the first review appeared in 1959 (Dische and Benfield, 1959). Since that date, many additional cases have been reported. Owen *et al.* in 1964 gathered only 60 sufficiently proven cases. In 1972, we had gathered 87 fairly proven cases. Subsequently, two other reviews were published in 1981 and 1984, which brought up-to-date information contained in the preview ones (Girolami *et al.*, 1984; Ragni *et al.*, 1981). Several additional cases have been reported in the past two decades, together with molecular investigation of the defect, and the database is always expanding (McVey *et al.*, 2001). Probably, several cases went unreported, since they were deemed to add very little, if any, to the knowledge of the disease. These figures indicate that the defect is still rare but is definitely more frequent than Factor X or Factor II deficiency. Its frequency can be estimated around 1/300,000 people, definitely higher even than that for Factor V deficiency.

Recent reviews appeared in 1995, 1998, and 2002 mainly dealt with molecular biology changes or with both congenital and acquired alterations forms (Cooper *et al.*, 1997a; Iannello *et al.*, 1998; Perry, 2002; Tuddenham *et al.*, 1995). Chapters dealing with FVII published in recent editions of widely used books are sometimes not completely satisfactory because information is scattered in different chapters.

The purpose of this chapter is to combine clinical, genetic, laboratory, and molecular biology data dealing with FVII deficiency and abnormalities in a unified presentation.

A. Classification and prevalence

Until a few years ago, it was thought that Factor VII deficiency represented a single entity in which a defective Factor VII synthesis occurred. The first demonstration of the presence of an excess of CRM over the clotting counterparts was given by Goodnight *et al.* (1971). Since then other reports have confirmed the findings. In some instances, the FVII protein level was reported normal. It was then concluded that Factor VII defect could be subdivided into three major variants, as namely, Factor VIIneg (CRM negative), Factor VIIred (CRM reduced), and Factor VIIpos (CRM positive) (Cooper *et al.*, 1997a; Goodnight *et al.*, 1971; Mazzucconi *et al.*, 1977). These findings are in agreement with what demonstrated for prothrombin, for Factor IX and Factor X.

A classification based only on the level of activity and antigen is untenable, since it does not take into account the different reactivity toward different tissue thromboplastins shown by some of these abnormal FVII. In fact, it was demonstrated in 1978 that an abnormal FVII, namely, a Factor VIIpos variant, may either be inert toward some tissue thromboplastin or be normally activated by others (Girolami *et al.*, 1978a, 1979; Triplett *et al.*, 1985); therefore, different levels of FVII can be obtained using different tissue thromboplastins. This discovery has radically changed the approach toward the classification of FVII defects. The correct approach is the following:

1. Factor VII "true" deficiency (homozygous and heterozygous)
2. Factor VII abnormalities (homozygous, heterozygous, and compound or double heterozygous)
   a. Forms with discrepancy between clotting and antigen levels
   b. Forms not only with the same discrepancy but also with different behavior toward tissue thromboplastins of different origin.
3. Combined Factor VII defects with other clotting factors

Factor VII CRMneg are those cases with no detectable or very low antigen and therefore no activity (<10%). Factor VIIred are those with reduced FVII antigen (usually 20–50% of normal). However, such levels are always definitely higher than the clotting counterparts. Finally, Factor VIIpos are those cases in which FVII activity level is always low and FVII antigen is normal. The exact collocation of Factor VIIred may be difficult, since sometimes a similar pattern (intermediate FVII antigen and low FVII activity) may be due to the presence of a compound or double heterozygosis (hypo-dys form). Eponymous should be applied only to abnormalities, CRMpos.

The exact incidence of CRMpos variants has not been surely established yet (Girolami *et al.*, 1979, 1984; Goodnight *et al.*, 1971; Mazzucconi *et al.*, 1977). The overall picture would suggest that about 50% of patients show no FVII antigen in their plasma, namely, are CRM negative. This is a little less than that described for hemophilia B where about 60% of cases are B negative. Using another approach, FVII defects with low but parallel levels of FVII activity and antigen may be termed Type I (cases of true deficiency), and those with low activity and normal or reduced antigen may be defined as Type II.

As far as mechanisms or function are concerned, FVII deficiency may be subdivided into: (1) defects in secretion, (2) defects in binding to tissue factor (TF), (3) defects in carboxylation, and (4) defects in Factor X activation.

Molecular biology studies are supplying useful information for a satisfactory classification. However, the picture is not clear yet. The same point mutation has been described in patients who showed a different clotting and antigenic pattern. On the contrary, a similar clotting and antigenic pattern has been described in patients with different mutations (Arbini *et al.*, 1994; Bernardi *et al.*, 1993, 1994; Chaing *et al.*, 1994; Fromovich-Amit *et al.*, 2004; Gamba *et al.*, 1998; Gomez *et al.*, 2004; James *et al.*, 1993a; James *et al.*, 1993b; James *et al.*, 1997; Kuppuswamy *et al.*, 1993; Mariani *et al.*, 2005; Peyvandi *et al.*, 2000; Shurafa *et al.*, personal communication; Wu *et al.*, 2006; Wulff and Herrmann, 2000).

B. Biochemistry

Factor VII is one of the PC factors. It is synthesized in the liver in the presence of vitamin K together with Factors II, IX, and X and Protein C and Protein S.

It is a single polypeptide glycoprotein of a molecular weight of about 50,000 Da.

Like the other PC factors, Factor VII has in its COOH-terminal region a number of glutamic acid residues that are, in the presence of vitamin K, converted to γ-carboxyglutamic residues. The plasma concentration of FVII is about 0.1 mg%, namely, about 1/10 the concentration of Factor X and about 1/20 the concentration of prothrombin. This very low concentration explains why FVII could not be easily demonstrated by precipitating techniques such as by electroimmunoassay, immunodiffusion, or radial immunodiffusion. It has a short half-life of about 4–6 h.

Although FVII appears to be an active enzyme, its activity is increased about 40-fold when converted to its active form by Factors Xa, IXa, and XIIa or through Factor VIIa. Factor VIIa is a two-chain serine protease of 26,000 and 22,000 Da, held together by disulfide bonds.

The activation of factor VII occurs after binding to TF by cleavage of a single bond (Arg152–Ile153). Activated FVII requires TF and calcium to show significant proteolytic activity toward its substrates, namely, Factors IX and X.

FVII, similarly to other vitamin K-dependent clotting factors, contains several domains, namely, a Gla domain, epidermal growth factor (EGF)-like or kringle domains and a C-terminal catalytic domain homologous to trypsin.

FVIIa by itself has only a weak enzymatic activity and has a half-life similar or even shorter than that of the zymogen.

FVII seems to be the only known ligand to TF, and the complex formation is regulated by a specific inhibitor, the TF pathway inhibitor (TFPI), which forms a quaternary complex involving FVII and both the TF–FVIIa complex and Factor X. The role of TFPI in Factor VII deficiencies has not been extensively studied yet.

The heavy chain of FVIIa contains the catalytic domain, whereas the light chain of FVIIa contains the γ-carboxyglutamic acid (Gla) residues involved in calcium and membrane phospholipid binding. FVIIa is normally present in the circulation at trace levels, and it is maintained to serve at the physiological initiator of coagulation.

C. Genetics and hereditary transmission

The hereditary transmission of Factor VII deficiency is autosomal recessive or incompletely recessive (Alexander *et al.*, 1951; Cooper *et al.*, 1997a; Girolami *et al.*, 1984). Consanguinity is present in about 30% of families. The FVII level usually found in homozygous patients with cause deficiencies is below 10% of normal. By definition, all these patients have also low or undetectable levels of FVII antigen. This hereditary pattern seems to be present even in the case of abnormalities (Factor VII Padua, for example). In this regard, it is important to remember that, as seen for other PC factors, an abnormality may be associated with a ''true'' deficiency or with a different abnormality of the same factor (double or compound heterozygotes).

Such combined or double heterozygotes may represent a difficult diagnostic problem, and the evaluation of the antigen level is crucial in the understanding of the hereditary pattern. Parents or children of homozygous patients are obligatory heterozygous. The FVII level usually found in these heterozygous patients varies from 40% to 60% of normal.

Molecular biology analysis may be necessary to clarify the hereditary pattern in some families. However, because of their technical difficulties these studies have no widespread use in routine clinical and laboratory practice.

The human Factor VII gene is located on the long arm of chromosome 13 (q34), near the Factor X gene. It is composed of 9 exons and 8 introns with an overall organization similar to Factor IX, Factor X, and Protein C.

Exons 1 and 2 seem involved in a single sequence; Exon 2 encodes the propeptide and the Gln domain; Exon 3 is responsible for a hydrophobic region; Exons 4 and 5 control the EGF domains; Exons 6 and 7 control the activation process; and, finally, Exon 8 encodes the catalytic domain (Tables 14.11 and 14.12).

At present, there are about 300 patients who have been investigated from a molecular point of view. Several of these mutations are identical, even though sometimes the phenotype appears different. For example, there are at least five families with FVII deficiency (Padua, Little Rock, Detroit, Richmond, and Verona) due to the same mutation, namely, the Arg304Gln in Exon 8. This may create confusion and should be avoided. The first family described should be kept as the index case. Minor laboratory

Table 14.11 Selected and representative examples of kindreds with true deficiency (Type I)

References	Activity	Antigen	Clinical picture	Molecular defect	Functional defect	Genotype
Tuddenham *et al.* (1995)	1.7	1.2	Severe	Del C10696 Del C10785	Catalytic	Comp. Het.
Arbini *et al.* (1994)	<2	<4	Moderate	Ala294Val Del C11125	Catalytic	Comp. Het.
Peyvandi *et al.* (2000); (Kindred B1)	<1	10	Severe	Gln100Arg	EGF domain	Hom.
Wulff and Herrmann (2000)	8	4	?	G64A	Pro-peptide	Hom.
Fromovich-Amit *et al.* (2004)	5.5	6	Mild	Ala244Val	Catalytic	Comp. Het.
Gomez *et al.* (2004)	0.4	1	Severe	Gly117Arg Arg152 X	Secretion Stability	Comp. Het.

Comp. Het., compound heterozygotes; Hom., homozygotes.

Table 14.12 Selected and representative examples of genetic variants of Factor VII defects with normal or near normal antigen (CRM positive or Type II forms)

Variant	Activity (% of normal)	Antigen	Clinical picture	Molecular defect	Functional defect	Genotype
Kansas (Kuppuswamy *et al.*, 1993)	4	60%	Asymptomatic	Gln100Arg	Defective binding	Comp. Het.
Padua (James *et al.*, 1993b)	9 (*)	100%	Moderate	Arg304Gln	Impaired activation	Hom.
Charlotte (Chaing *et al.* (1994)	<1	100%	Severe	Arg152Gln + Arg79Gln	Decreased binding to TF + impaired activation	Double Hom.
Kindred H (Peyvandi *et al.* (2000)	<1	135%	Severe	Pro303Thr	Catalytic domain	Hom.
No eponym (Fromovich-Amit *et al.* (2004))	<1	83%	Mild	Cys310Phe	Catalytic domain	Hom.

Only homozygotes or compound heterozygotes have been taken into consideration.
* Normal with ox-brain thromboplastin; Comp. Het., compound heterozygotes; Hom., homozygotes.

Table 14.13 Different behavior in clotting and immunological assays of Factor VII abnormalities due to the same amino acid substitution (Arg304Gln)

	Factor VII activity (% of normal)			
Defect	Rabbit-brain	Human brain or placenta	Ox-brain	Factor VII antigen (% of normal)
FVII Padua	8	30	100	100
FVII Verona	20	20	50	60
FVII Little Rock	7	16	30	67
FVII Detroit (*)	14	21	124	80
FVII Richmond	n.r.	14	n.r.	120

Differences may be explained either by different procedures or by the presence of additional, unknown defects.
* Pattern is more close to that observed in FVII Padua; n.r., not reported.

differences seen in these patients may be either due to the different laboratory techniques or due to the presence of other unknown molecular changes (Table 14.13). The same mutation was first seen in an isolated patient by O'Brien *et al.*, but no family study or extensive clotting investigation was supplied (O'Brien *et al.*, 1991).

By eliminating the duplets, one reaches the approximate number of about 200. These unique alterations include missense mutations (the large majority), splice junction and promoter changes (rare) and nonsense mutation, deletion or insertions (very rare). All areas of the gene are involved (promoter, Gla domain, etc.). The large majority involve Exon 8. Several areas or domains seem involved in the binding to TFs, and this may explain the different results observed sometimes using thromboplastins of different origin. As far as FVII activation is concerned, residues between 300 and 330 seem critical. Several substitutions in this area of Exon 8 have been associated with low activity and normal antigen (CRM positive). Furthermore, peptide inhibition studies have shown that residues 330–340 are important for both Factor IX and Factor X binding and activation (James *et al.*, 1997).

On the contrary, substitutions on Exons 1–7 are more frequently associated with cases of true deficiency (CRMneg). Recent studies have indicated that the Cys329 residue plays a critical role in FVII function (Wu *et al.*, 2006).

A large number of molecular studies have not been as rewarding as originally expected. It is becoming every day more evident that different mutations may give the same or similar phenotype and, conversely, that different phenotypes may be caused by the same mutation (ex Arg304Gln).

From a clinical point of view, a complete and accurate evaluation of FVII levels activity and antigen appears sufficient.

D. Clinical picture

There is no typical clinical feature for Factor VII defect. Only homozygotes are variably symptomatic. Heterozygotes are always asymptomatic contrary to what is showed for heterozygotes with Factor II and Factor X deficiency (Table 14.14). The wide clinical spectrum of hemorrhagic manifestations justified the fact that in some cases a minor bleeding is maintained to be present; in others, a major hemorrhagic condition is described. Such variability seems to correlate only partially with FVII levels, which in homozygous patient is usually below 10% of normal (Giansily-Blaizot and Schved, 2005). The discrepancy seen in some patients between FVII levels and bleeding tendency remains to be clarified. It has to be observed that this discrepancy may sometimes be more apparent than real because of the different levels of FVII activity obtained using tissue thromboplastins of different origin. If comparison is not made with reagents that yields the lowest FVII activity level, wrong conclusions could be drawn. Bleeding tendency is characterized by easy bruising, gastrointestinal bleeding, hematuria, epistaxis, and, in several cases, hemarthrosis, occasionally with severe orthopedic impairment. Menometrorrhagias may be present and severe in women. Oral contraceptives have been shown to decrease this type of bleeding. An increased incidence of cerebrovascular bleeding has also been demonstrated. This is often the cause of death or serious neurological sequelae. As said before, there is nothing typical in this bleeding tendency. Note that spontaneous or, more frequently, replacement therapy-associated thromboembolic manifestations have been described in FVII deficiency, indicating that a FVII defect is unable to fully protect from thromboembolic phenomena (Girolami *et al.*, 1978a; Sabharwal *et al.*, 1992). Pregnancy has been brought to term under adequate substitution therapy.

E. Laboratory diagnosis

The main criteria for the diagnosis of Factor VII deficiency are: prolonged prothrombin time corrected by the addition of normal plasma or serum but not by the addition of adsorbed plasma, and normal partial thromboplastin and thrombin time. In the past, it was maintained that a prolonged prothrombin time together with a normal Roussell viper venom (RVV) clotting time was sufficient for a sure diagnosis. This has been demonstrated not to be the case, since a Factor X abnormality, Factor X Friuli, also shows a similar pattern (Table 14.15). In the latter case, however, a prolonged PTT is always present. Furthermore, there are rare families with Factor X deficiencies, which appear defective only in the extrinsic system. As a consequence, the pattern obtained is identical to that seen in FVII deficiency, namely, normal PTT, prolonged PT, and normal TT.

Table 14.14 Comparison in incidence of minor bleeding manifestations between heterozygotes for Factor II or Factor VII or Factor X deficiency and hemophilia B carriers and unaffected family members as seen in Padua

Defect	No. affected	Symptomatic No. (%)	No. of nonaffected	Symptomatic No. (%)	Comments
Factor II	26	9 (34.6)	31	1 (3.2)	Epistaxis, easy bruising, menhorragia, excessive bleeding after tooth extraction or surgery
Factor VII	60	2 (2.4)	52	2 (3.8)	
Factor IX (hemophilia carrier)	38	1 (2.6)	45	2 (4.4)	
Factor X	120	36 (30)	81	2 (2.4)	

The difference is significant only for Factors II and X defects.

Table 14.15 Laboratory differential diagnosis between Factors VII and X deficiencies and some variants

Defect	PTT	PT rabbit-brain/ ox-brain	RVV	Antigen (CRM)	Comments
Factor VII true deficiency	N	P/P	N	No	
Factor VII Variant VII Padua	N	P/N	N	Yes	Other variants may have different pattern
Factor X true deficiency	P	P/P	P	No	
Factor X Variant X-Friuli	P	P/P	N	Yes	Other variants may have different pattern
Factor X Variant X-Padua	N	P/P	N	Yes	Other variants may have different pattern

CRM, cross-reacting material; RVV, Roussell viper venom; P, prolonged; N, normal.

Prothrombin time derivate tests, such as PP test, Normotest, and Thrombotest, are all variably prolonged. FVII activity is low, as expected. FVII activity levels are usually below 10% of normal in homozygote patients and between 40% and 60% of normal in the heterozygote population.

More variable levels may be encountered in compound heterozygotes, especially in dys-dys forms. Therefore, a sure allocation of heterozygotes may be complicate in the case of compound or double heterozygosity. In this case, the independent segregation of the two defects should be encountered in the family members. Factor VII assays can be carried out as follows: (1) using a tissue thromboplastin as activator in a one-stage procedure, (2) by means of chromogenic substrates, (3) by neutralization test using anti-FVII antiserum, (4) by an ELISA method, and (5) by radioimmunoassay.

Only the first one is widely used. It is important to use at least two tissue thromboplastins (e.g., rabbit-brain and ox-brain thromboplatins). If no major discrepancy is present in the FVII levels so obtained, the defect is probably a "true" FVII deficiency. If a discrepancy over 15% in the FVII level is present between the two clotting assays, a FVII abnormality should be suspected. In this case, thromboplastins of other origin, human for example, should also be used. The substrate best suited for such assay is represented by fresh or lyophilized congenital FVII deficient plasma.

If lyophilized or stored plasma is used, a mixture of such plasma with absorbed normal plasma is preferable to assure adequate Factor V levels in the system. Artificially prepared substrates are not satisfactory. The routine use of chromogenic assays is not practical so far, since the method requires the presence of purified Factor X or a three-stage rather complicated assay (Margaritella and Deutsch, 1981; Seligsohn *et al*., 1978). Furthermore, it is more complicated and expensive than the plain clotting test. In the characterization of an abnormality, it is imperative to carry out also an immunological assay. The neutralization test was widely used in the past often with equivocal or doubtful results.

The neutralization test is based on adding antiserum to the test plasma (first mixture or stage I). Subsequently, after 1 h incubation at 37 °C, an aliquot is mixed with equal parts of pooled normal plasma (second mixture or stage II). Finally, FVII is assayed in this second mixture. The residual FVII activity found in the second mixture is inversely related to the presence of FVII protein in the test plasma. In the evaluation of FVII abnormalities, it is important to use in the assay of residual FVII activity (stage III), the thromboplastin that yields the lowest activity (Girolami *et al*., 1984). The radioimmunoassay is surely more precise but has the limitation of the need of I^{125} labeled FVII. The availability of suited precipitating antisera has simplified the matter allowing the performance of, for example, an electroimmunoassay or a construction of an ELISA. The ELISA method has also been introduced in recent years and has acquired widespread use. It seems superior and less time consuming than the neutralization test and the radioimmunoassay (Fair, 1983).

The correct identification of the heterozygotes is usually fairly simple, but in some patients with borderline FVII activity values (60–70% of normal) the procedure may be difficult. In our hands, the best approach to that has been the accurate and repeated FVII activity assays on different occasions. In heterozygotes, the level will be found constantly even though only slightly decreased. Patients with acquired conditions tend to present variable levels. Furthermore, the administration of vitamin K will usually correct the defect only in acquired conditions. Finally, in acquired defects there is always a concomitant decrease in other factors, usually Factors II, IX, and X.

The evaluation of activated Factor VII (FVIIa) has been proposed, but its value, in clinical practice, remains to be proven (Morrisey *et al*., 1993). The principle of the test is based on the discrepant sensitivity of a recombinant soluble TF toward FVII and FVIIa. This particular TF (truncated TF) does not activate FVII into FVIIa and therefore is maintained to be independent of the FVII present in the plasma. In other words, it measures only FVIIa already existing in the plasma. Normally about 99% of FVII circulating in plasma is the zymogen form and about 1% circulates in the active form. Molecular biology techniques, inclusive occasionally of expression studies, have been carried out in the most recent era. This has greatly contributed to the characterization of several variants (Fig. 14.3).

Of paramount importance in the diagnosis of congenital FVII defects, regardless of the type, are the following criteria: (1) fixity of the defect; (2) no influence by vitamin K administration but for the case of combined Factors II, VII, IX, and X; and (3) the presence of similar defects in the family.

F. Prognosis

The prognosis of congenital FVII deficiency is guarded. The life expectancy varies considerably and is in a fair relationship with the FVII activity level. The improved therapeutic approach due to the availability of FVII concentrates is having a positive effect on survival. However, no study is available on the survival of these patients yet.

Cerebral hemorrhages seem a frequent cause of death in patients with very low-activity levels (Ragni *et al.*, 1981). On the contrary, other patients with FVII activity of about 10% of normal lead an almost normal life (. . .). Heterozygous have an excellent prognosis, since they are always asymptomatic.

G. Therapy

Factor VII has a short survival half-life, about 5 h, and therefore substitution therapy may be a problem in the presence of a severe bleeding emergency (traumas, surgery, cerebrovascular bleeding, etc.). The administration of fresh frozen plasma may not be practical because of the low yield. Prothrombin complex concentrates, titrated for Factor IX, contain variable amounts of FVII. In our hands, the use of such concentrates resulted in satisfactory FVII levels (Fig. 14.4), but it entailed in the past a very high incidence of nonA-nonB hepatitis. Approximately 1 of 2 patients developed clinical and/or laboratory evidence of hepatitis. The pattern has improved in recent years with better sterilization procedures. Safe therapeutic levels of FVII seem to be around 30–40% of normal. This is in agreement with the known fact that heterozygotes, which have about 50% activity levels, do not show bleeding manifestations. Isolated Factor VII concentrates, recombinant or not, are also available. The effect of antifibrinolytic agents on survival times has not been investigated. On the basis of the experience gathered from other clotting factors, it could be claimed that the association of such compounds together with substitution therapy may be beneficial.

During the past decade, a concentrate of recombinant-activated FVII has been widely used in the treatment of Factor VII deficiency (Brenner and Wiis, 2007). We think this is not always justified both for reasons of costs and for the danger of thrombotic complication which, even though rarely, do occur (O'Connell *et al.*, 2006). Furthermore, there is no

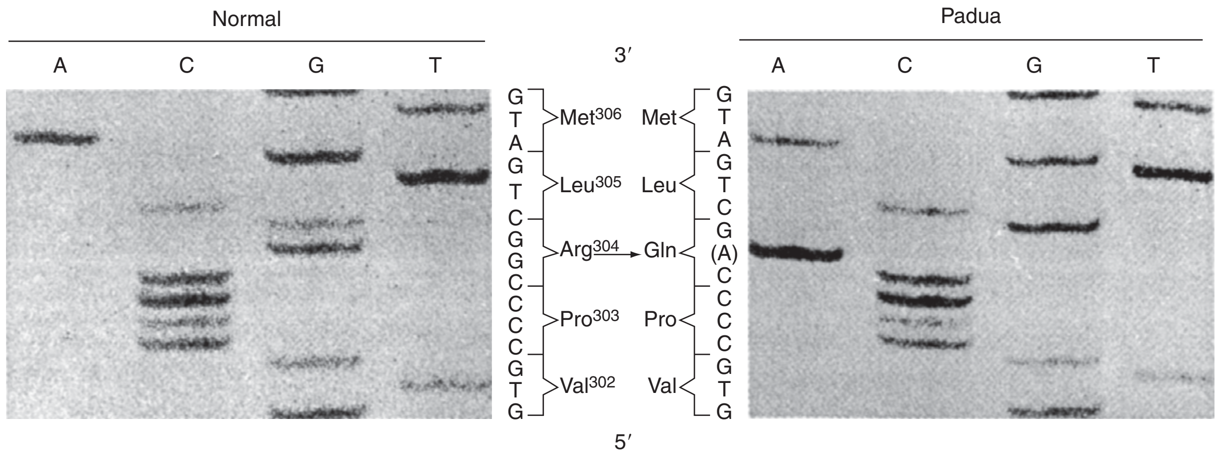

Figure 14.3 Autoradiography showing the point mutation at residue 10,828 in Exon VIII of FVII Padua gene. The base substitution is shown in parentheses. The mutation results in coding for glutamine instead of arginine at position 304 of the abnormal FVII molecule.

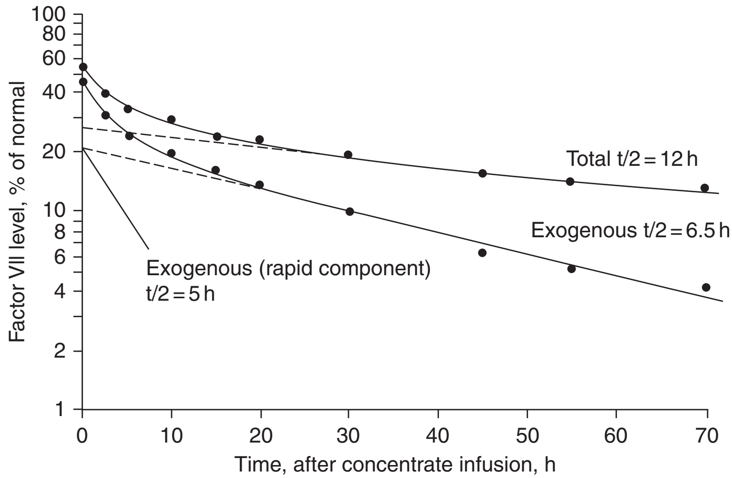

Figure 14.4 **This FVII deficient patient received 1,000 units of Factor VII concentrate together with 300 ml of fresh plasma (about 15 units/kg body weight) in 30 min. The survival time of exogenous component was 6.5 h.**

available demonstration of the superiority of this concentrate with regard to the standard FVII concentrates. Oral contraceptives have been shown to decreased menorrhagia and improved anemia and also to prevent ovulation-induced hemoperitoneum (Girolami *et al.*, 2006; Payne *et al.*, 2007). For the sake of completion, it has to be remembered that orthotopic liver transplant is a potential therapeutic option. It has been reported so far in four cases with apparent good results (Dhawan *et al.*, 2004; Levi *et al.*, 2001). Two of such patients, before transplantation, had undergone injection of hepatocytes in the portal system with only temporary and partial increase of FVII level (Dhawan *et al.*, 2004). On the contrary, no data are available with regard to attempts of gene therapy.

H. Association with other clotting and non-clotting defects

In addition to the bleeding condition due to the combined defect of all vitamin K-dependent clotting factors, which will be dealt with separately, occasional reports of the association of FVII with other clotting factors have been reported in a given patient and in his or her family.

These associations are important because they could create diagnostic difficulties.

These combined defects may be subdivided for simplicity in: (a) association with one of the other PC factors, (b) association with non-vitamin K-dependent factors, and (c) association with non-clotting abnormalities.

1. Association with other prothrombin complex factors

Several reports have dealt with families in which the concomitant presence of two of these factors was present. Two single PC factors have been reported in association with FVII deficiency, namely, Factor IX and Factor X.

The association with Factor IX, combined defect FVII + FIX, has received considerable attention in the past. However, the condition has never been fully understood and accepted.

In some families, there seem to be an independent segregation of the two defects, namely, hemophilia B and heterozygous FVII deficiency (Hall *et al.*, 1975). In some instances, the combined defect has been maintained to be due to a laboratory artifact (low FVII level in hemophilia B plasmas, especially hemophilia BM plasmas).

Unfortunately, no molecular studies in these patients have been carried out, and therefore no final conclusion can be drawn as to the significance of this combined defect.

The second condition, namely, combined FVII and Factor X deficiency, apparently consists of two forms. In the first form (Type I), there is a casual association of the two defects in the same family. At least, few families have been described. In these families, there is an independent segregation of the two defects (Boxus *et al.*, 1997; Kroll *et al.*, 1964; Menegatti *et al.*, 2004).

However, another form of combined FVII and Factor X defect has been reported. This form is due to alteration in chromosome 13 (q.34) where both FVII and Factor X genes are located (Ott and Pfeiffer, 1984). In this case, there is no independent segregation of the two defects in the family, and the hereditary pattern is autosomal dominant (Ott and Pfeiffer, 1984; Pfeiffer *et al.*, 1982).

Often, these patients have also malformations and defects unrelated to blood coagulation such as mental retardation and bone abnormalities.

2. Association with non-vitamin K-dependent factors

Other associated coagulative defects have been reported with Factor VIII or Factor V deficiencies (Ambriz-Fernandez *et al.*, 1985; Gaston *et al.*, 1961; Girolami *et al.*, 1977b, 2007; Machin and Miller, 1980).

The significance of these combined defects remains to be clarified. It seems that two types of combined Factors VII and VIII exist (Girolami *et al.*, 1977b). The first type consists in the chance association of heterozygous Factor VII deficiency with hemophilia A. The second type of combined defects is more interesting in the sense that it seems to be secondary to a common alteration involving both factors. However, this has not been proven.

A few families with combined Factor VII and Factor V defects have been studied (Ambriz-Fernandez *et al.*, 1985; Girolami *et al.*, 2007; Rotoli *et al.*, 1983). One of these cases has been investigated by molecular biology techniques and confirmed to be due to casual association between defects independently derived from parents (Traynis *et al.*, 2006).

Occasional association with von Willebrand disease has also been reported. In this case, an independent segregation of the two defects noted in the families has been demonstrated (Girolami *et al.*, 2007). Other associations were described with Factor XI and Factor XII (Slease and Schumacher, 1977; Berube *et al.*, 1992).

Because of the relative diffusion of the defect, it is of no surprise that these combined defects have been described. Even an association with Protein C deficiency has been described (Takeuchi *et al.*, 1988). Interestingly, this patient had pulmonary embolism.

3. Association with non-coagulative abnormalities

The other diseases associated with FVII defect were some bilirubin metabolism diseases such as Gilbert or Dubin–Johnson (D–J) (Seligsohn *et al.*, 1969, 1970a). These defects, particularly in D–J, appear frequently in certain Jews populations, and the association with FVII defect may represent the result of an independent hereditary transmission (Seligsohn *et al.*, 1970b). Recent investigations have shown that one of the mutations involving the multi-organ resistance Protein Z transporter causing D–J, namely, the I1173F, was present in Iranian Jewish, whereas another, the R1150H, was present in Moroccan Jews. Since both mutations were found associated with FVII deficiency due to the same A244V missense mutation, it was concluded that the D–J mutations occurred later in time, namely, after the Jewish migration to these countries, about 600 BC (Mor-Cohen *et al.*, 2007).

IV. Congenital Factor IX Deficiency (Hemophilia B)

The most astonishing fact about Factor IX (FIX) deficiency is its vitamin K dependency but its sex-link hereditary transmission, whereas all other vitamin K-dependent factors have an autosomal pattern of inheritance.

The isolated deficiency of FIX is termed hemophilia B.

This defect was first recognized because of the observation by Pavlovsky in the late 1940s that there was a correction of the defect when the plasma of two ‘‘hemophilic’’ patients was mixed (Pavlovsky, 1947).

A few years later, the distinction was clearly established by Aggeler *et al.* and Biggs *et al.* (Aggeler *et al.*, 1952; Biggs *et al.*, 1952).

Hemophilia B is less frequent than hemophilia A, with an approximate a 1/8 or 1/10 ratio.

The hereditary transmission is sex linked. The FIX gene is located on the X chromosome not far from the gene controlling Factor VIII activation. Both sporadic and familial forms have been described.

The clinical picture is indistinguishable from that seen in hemophilia A. Hemarthroses, deep muscle hematomas, hematuria, and easy bruising are most frequent manifestations.

A. Classification and prevalence

Classification: the defect may be subdivided in severe (FIX activity <1% of normal), moderate (FIX activity, 1–5% of normal), and mild (FIX activity, 5–35% of normal). Mild forms may be asymptomatic; moderate forms usually bleed after surgery or trauma. Severe forms show spontaneous bleeding.

Contrary to what seen in hemophilia A, antigen levels may be absent (CRMneg), reduced (CRMred), or normal (CRMpos) (Denson *et al.*, 1968). Several variants have been described. The most important ones are FIX Leiden and FIX Bm. About 60% of patients show no CRM in their plasma. About 30% of the remaining 40% are of the Bm type.

The main feature of hemophilia B Leiden is a tendency to improve with age. The main feature of hemophilia Bm is to show an inhibitory effect on the FVII-tissue thromboplastin reaction when the latter is obtained from ox-brain or pig-brain. In other words, the Bm hemophilia behaves as *in vitro* "anticoagulant" or "inhibitor" in the prothrombin time system.

Recent molecular biology studies have supplied a lot of information, and now the number of mutations responsible for hemophilia B is countless.

In the past, another form of hemophilia B received some attention, namely, the form associated with mild Factor VII deficiency (Elodi, 1973; Elodi and Puskas, 1972; Girolami *et al.*, 1977a). However, the existence of such form has not been fully confirmed (Girolami *et al.*, 2007).

B. Hereditary transmission and genetics

The hereditary transmission is X-linked as hemophilia A. The carrier state is characterized by slightly decreased or borderline levels of FIX activity with normal or near normal FIX antigen.

The gene for FIX is located in the long arm of X chromosome, close to the Factor VIII gene.

It consists of eight exons and seven introns. Exons I and II encode for a pre-pro leader sequence and the Gla domain. Exon III, which is a very short one, is involved in the aromatic stock domain and the last γ-carboxyglutamic acid residue. Exons IV and V encode the EGF domain. Exons VI, VII, and VIII encode the activation peptide and the catalytic domain.

The number of chromosomal alterations reported in hemophilia B is very large, and therefore it is impossible to go into details. Tuddenham and

Cooper listed about 150 samples (Tuddenham and Cooper, 1994). Several hundred more have been added subsequently, bringing the total now to about 2,500 (database).

Briefly, mutations may be subdivided into: (1) deletions, (2) point mutations (both missense and nonsense), (3) insertions, and (4) inversions. The majority of missense mutations found in mild hemophilia B are not sporadic or *de novo*. On the contrary, the majority of mutations found in the moderate and severe hemophilia B are of the sporadic or the *de novo* type. A representative selection of these molecular alterations have been gathered in Tables 14.16 and 14.17.

All regions of FIX gene have been involved. Several mutations, localized in different areas, have given origin to severe hemophilia B. Among those involving the promoter region, FIX Leiden is worth of special mutation. The patient with FIX Leiden shows a tendency to improve with age. The defect consists in a single nucleotide substitution in an area of the promoter region (usually −21) and (Bezeaud *et al.*, 1984) sensitive to androgens (androgen responsive element or ARE) (Crossley *et al.*, 1992; Reijnen *et al.*, 1992, 1993; Reitsma *et al.*, 1988). Similar hemophilia B features have been described even with mutations in a wider area (−21 + 13 region) (Giannelli *et al.*, 1998, 1999; Triplett *et al.*, 1985). Astonishingly, mutations at −26 seem to disrupt completely the ARE, and therefore the hemophilia B patients carrying this mutation fail to improve with age. The promoter mutations responsible for hemophilia Bm variably alter the pattern of expression, whereby the FIX gene becomes transcriptionally

Table 14.16 Selected examples of mutations seen in some cases of hemophilia B without antigen (CRM negative)

References	Eponym n.r. or omitted (see text)	Mutation (Exon)	Phenotype
Ludwig *et al.* (1989)	–	Arg338 Stop (Exon VIII)	Severe
Attree *et al.* (1989)	–	Lys411 Stop (Exon VIII)	Severe
Green *et al.* (1989)	–	Asn120Tyr (Exon V)	Severe
Koeberl *et al.* (1990)	–	Gly60Ser (Exon IV)	Mild
Chen *et al.* (1991)	–	Trp215 Stop (Exon VII)	Severe
Winship and Dragon (1991)	–	Glu7Ala (Exon II)	Mild

This pattern may be seen with mutations in any area; n.r., not reported.

Table 14.17 Selected examples of mutations seen in some cases of hemophilia B with normal FIX antigen (CRM positive)

References	Eponym	Mutation (Exon)	Phenotype
Noyes *et al.* (1983)	Chapel Hill	Arg145Hys (Exon VI)	Mild
Davis *et al.* (1987)	Alabama	Asp47Gly (Exon IV)	Mild phenotype
Reitsma *et al.* (1988)	Leiden	Promoter	Androgen sensitive
Tsang *et al.* (1988)	London 2	Arg333Gln (Exon VIII)	Severe
Ware *et al.* (1989)	San Dimas	Arg -4 Gln (Exon II)	Severe phenotype
Bertina *et al.* (1990)	Novara	Arg180Gln (Exon V)	Bm type
Lozier *et al.* (1990)	New London	Glu50Pro (Exon IV)	Mild
Chen *et al.* (1991)	n.r.	Val107Ala (Exon V)	Mild

Both mild and severe phenotypes have been described. Patients with mutations in Exon VII apparently do not present normal or near normal antigen. On the contrary, mutations in Exons II or VI show frequently CRM. CRM, cross-reacting material.

active only after puberty (Reijnen *et al.*, 1992; Reitsma *et al.*, 1988). The underlying mechanism for this behavior seems because of a defective binding of this mutated FIX promoter to hepatocyte nuclear factor-4 (HNF-4) (Reijnen *et al.*, 1993).

It would seem that androgen hormones may somehow mitigate or even neutralize the defect. These patients note, after puberty, a decrease of the frequency and intensity of bleeding manifestations; FIX levels increase and even severe hemophiliacs may be changed in mild ones.

Another peculiar form of FIX deficiency is the hemophilia Bm. This defect, first described by Hougie and Twomey (1967), is characterized by the presence of a CRM that interferes, at least *in vitro*, with the TF + Factor VII reaction when certain slow thromboplastins, for example ox-brain or pig-brain thromboplastin, are used in the test. It is estimated that about 15% of patients with CRM show this phenomenon. This condition may sometimes create diagnostic difficulties as will be discussed later. It has been demonstrated that the inhibitory effect correlates with the level of antigen. The maximal prolongation of ox-brain thromboplastin is observed in those patients who have normal or even slightly increased level of FIX antigen. Patients with lower levels of antigen show less inhibition (Girolami *et al.*, 1982a).

The genetic alterations seen in this variant involve Exon VI or Exon VIII. They are all point mutations. The reason why mutations in two nondependent exons may yield the same phenotype is unknown and confirms what seen also for other clotting factors, namely, that the same phenotype may be seen with different mutations occurring even in different regions. Furthermore, it has to be noted that there is no absolute correlation between mutation and phenotype. For example, the same mutation (Ile397Thr) has been seen both in families with and without the Bm phenomenon (Tuddenham and Cooper, 1994).

Point mutations are by far the most frequent alterations: they represent in fact about 90% of all the changes. Deletions are present in about 6–7% of cases, while insertions and other changes are very rare. This is quite different from what occurs in hemophilia A, where Intron 22 inversion is frequently present in severe forms of the disease (Bowen, 2002).

Besides the promoter region, changes in all exons (from Exon II to Exon VIII) have been associated with hemophilia B phenotype. Mutations in Exon VII seem to cause only cases of hemophilia B without any CRM (true deficiency). Mutations in Exon VIII also seem to cause a prevalence of CRMneg forms. Residues 390–397 of Exon VIII seem critical in binding to Factor X and/or to antithrombin. Selected but representative lists of mutations causing CRMneg or CRMpos hemophilia B are reported in Tables 14.16 and 14.17. A detailed analysis of different molecular alterations seen in FIX deficiency is beyond the scope of this clinical review that must deal mainly with the phenotype.

Hemophilia B has occasionally been described in females (Costa *et al.*, 2000; Di Paola *et al.*, 2004; Sellner and Price, 2005; Shetty *et al.*, 2001a; Spinelli *et al.*, 1976; Yang and Ragni, 2004). The molecular alterations seen in these patients are: (1) homozygosity for hemophilia gene (very rare), (2) nonrandom or skewed inactivity of X chromosome, (3) double *de novo* mutation, (4) mosaicism, (5) deletion of tip of long arm of X chromosome, (6) translocations, (7) segmental isodisomy, and (8) other structural abnormalities of X chromosome. Mosaicism may sometimes be associated with Turner's syndrome (45, X0).

Bleeding manifestations are usually less severe than those seen in classical hemophilia B patients.

FIX levels are variable, but very low levels as those seen in severe hemophilia B (around 1% of normal) are rare. Diagnosis may be difficult, since one does not expect to be facing a woman with hemophilia B.

C. Biochemistry and function

Factor IX is a single chain vitamin K-dependent glycoprotein with a molecular weight of about 57,000 Da.

It is constituted of 415 amino acids. Its structure is similar to that of other vitamin K-dependent factors. It contains several domains, namely, a 12 γ-carboxyglutamic acid residue domain located in the amino-terminal end; two EGF-like domains are also located in this part of the protein. Finally, there is an activation peptide together with the catalytic domain. It circulates at an approximate concentration of 0.4 mg%.

FIX is activated into FIXa by cleaving two arginine peptide bonds (Arg145 and Arg180). The result of such activation is the formation of a light chain (catalytic area) and a heavy chain held together by disulfide bonds together with an activation peptide. Activated FIX in the presence of phospholipids, Ca^{2+}, and Factor VIII forms a prothrombinase that activates Factor X to Factor Xa.

The main types of functional and/or biochemical defects at the base of hemophilia B could be summarized along the following lines: (1) defects with abnormal carboxylation, (2) cases with reduced activation from Factor XIa, (3) Bm type, (4) alteration in the catalytic area, and (5) changes in the androgen sensitive area (Attree *et al.*, 1989; Bachmann *et al.*, 1958; Bicocchi *et al.*, 2006; Chen *et al.*, 1991; Davis *et al.*, 1987; Green *et al.*, 1989; Koeberl *et al.*, 1990; Lozier *et al.*, 1990; Ludwig *et al.*, 1989, 1992; Mukherjee *et al.*, 2003; Noyes *et al.*, 1983; Odom *et al.*, 1994; Spitzer *et al.*, 1990; Taylor *et al.*, 1992; Tsang *et al.*, 1988; Usharani *et al.*, 1985; Ware *et al.*, 1989; Winship and Dragon, 1991).

D. Clinical picture

The clinical picture is similar to that of hemophilia A. No differentiation of the two forms is possible on clinical presentation.

The first clinical manifestation usually occurs in infancy because of ecchymosis and hematomas secondary to accidental falls. Hemarthroses may occur after a few years. These are posttraumatic at the beginning, later on they occur spontaneously. Patients with hemophilia B Leiden show an attenuation of the bleeding diathesis after puberty (Crossley *et al.*, 1992).

Easy bruising and superficial or deep muscle hematomas are frequent. As in hemophilia A, hematomas of the ileopsoas may occur. Hematuria is also frequent. Brain hemorrhage, as in other prothrombin factors deficiencies, may be the cause of death. Moderate forms with FIX levels between 2% and 5% are paucisymptomatic. Hemarthroses usually occur only after traumas. Mild forms (FIX level >5%) are usually asymptomatic. These patients usually bleed excessively only after surgery or trauma.

Carriers are asymptomatic. The same is true for mild forms that, unless challenged by trauma or surgery, may remain asymptomatic. The appearance of an inhibitor, which is approximately as frequent as in hemophilia A, aggravates the bleeding manifestations.

Women with hemophilia B are variably symptomatic according to the level of FIX. The cases due to skewed or nonrandom inactivation are usually less severe, equivalent to males with mild hemophilia and therefore show a limited bleeding tendency (Lusher and McMillan, 1978).

E. Diagnosis

There is no possibility to diagnose hemophilia B on clinical grounds. The laboratory criteria for the diagnosis of hemophilia B are: prolonged PTT, usually normal PT, and normal TT. FIX assay varies between 1% and 25% of normal, allowing the distinction between severe, moderate, and mild forms. There is a good correlation between FIX levels and clinical phenotype. The prolonged PTT is corrected by the addition of normal plasma, normal serum but not by the addition of adsorbed normal plasma. In the case of hemophilia Bm, prothrombin time is prolonged, particularly if one uses ox-brain thromboplastins such is the case with Thrombotest (Owren, 1959) (Table 14.18). The prolongation is variable but may be marked (ex 100 s vs. a normal of 40 s). Interestingly, such prolonged PT is not corrected by the addition of normal plasma, contrary to what happens for the PTT. As a matter of fact, in serial dilutions and cross-studies, it has been demonstrated that even 10–20 of the patient's plasma in a mixture with normal plasma is able to prolong slightly the ox-brain thromboplastin clotting time (Fig. 14.5). Furthermore, there is a correlation between the level of antigen or of the difference of antigen versus activity and the degree of the ox-brain thromboplastin prothrombin time (Fig. 14.6). Factor IX antigen is absent in about

Table 14.18 Behavior of clotting global tests in the differential diagnosis of Factor VII (FVII), Factor IX (FIX), and Factor X (FX) deficiencies and some variants

Defect	PTT	PT rabbit-brain	PT ox-brain	RVV	Comments
Classical FIX Def. (hemophilia B)	P	N	N	N	
Hemophilia Bm	P	N (1)	P	N	(1) It may be 1–2 s prolonged
Classical FX Def.	P	P	P	P	
FX Friuli	P	P	P	N	
FX Padua (Riadh)	N	P	P	N	
FX Melbourne	P	N	N	N	
Classical FVII Def.	N	P	P	N	
FVII Padua	N	P	N	N	

P, prolonged; N, normal; RVV, Roussell viper venom.

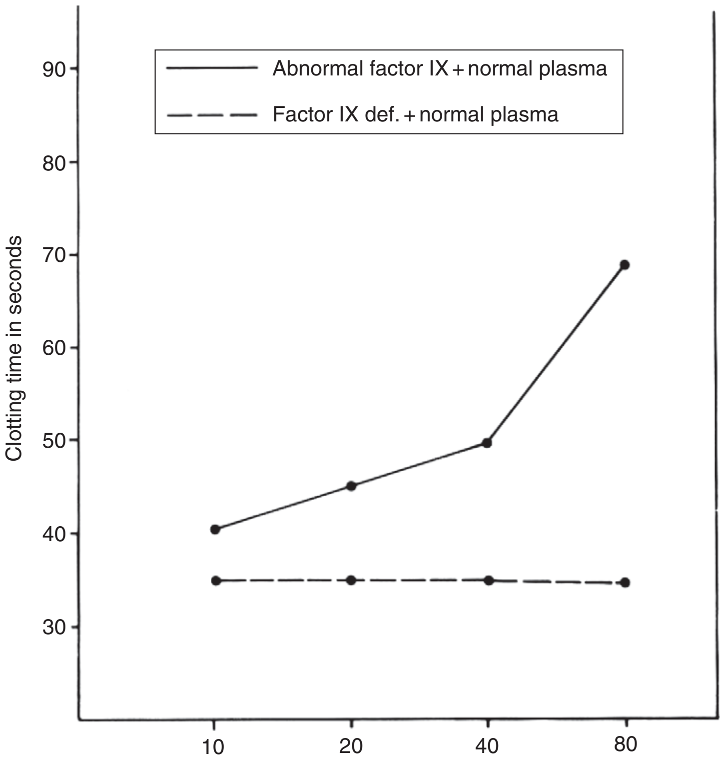

Figure 14.5 **Inhibitory effect of hemophilia Bm plasma on ox-brain thromboplastin prothrombin time (Thrombotest). Even the addition to normal plasma of 10–20% of hemophilia Bm plasma causes a prolongation of the Thrombotest. On the contrary, the addition to normal plasma of increasing aliquots of classical hemophilia B [cross-reacting material (CRM) negative] remains without effect. Percentile parts of patient plasma in abscissa.**

50% of patients. In the remaining 50%, it is variably present in a reduced or normal form. Cases with normal or near normal FIX antigen are about 30% of the total.

In case of hemophilia Bm (about 20% of cases with CRM), FIX antigen is always present even though to a variable extent (50–100% of normal). Carriers show borderline or slightly decrease FIX activity and, usually, normal FIX antigen.

Molecular biology analyses allow a more precise diagnosis but are not needed in routine clinical practice. The two most important variants, hemophilia B Leiden and hemophilia Bm, can be diagnosed by clinical and specific clotting test. Detection of carriers used to be carried out by clotting versus antigen evaluation. Recently, molecular biology techniques are widely used: DNA polymorphisms analysis, screening procedures such

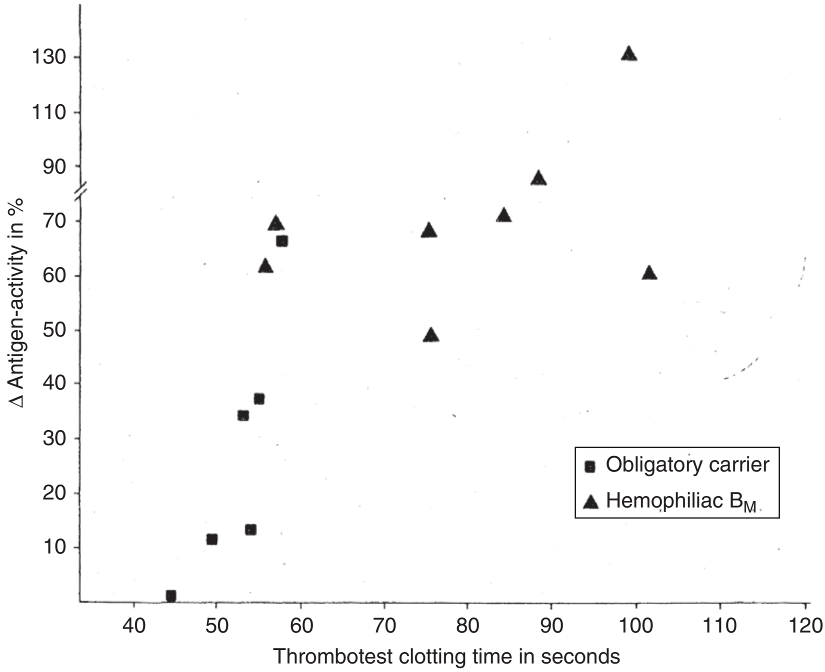

Figure 14.6 Correlation between Thrombotest clotting times (*abscissa*) and antigen–activity difference (*ordinate*). Besides the nine hemophilia Bm patients, available mothers or daughters of hemophiliacs (obligatory carriers) were also included. The mother of two known hemophiliacs, who were dead at the time of the study, is also included as an obligatory carrier. Note that the only patient with a normal Thrombotest had a Δ antigen–activity equal to zero. The correlation is significant ($r = +0.78$; $t = 4.30$; $p < 0.001$).

as conformation sensitive gel elettrophoresis (CSGE), and direct mutation detection (Ljung, 1969).

Prenatal diagnosis has been carried out by the same or similar techniques on material obtained by amniocentesis or chorionic villi biopsy. Factor IX assay on blood obtained by cordocentesis has also been useful in some cases (Shetty and Ghosh, 2007). Finally, direct analysis of fetal DNA circulating in maternal blood has been proposed (Costa *et al.*, 2002).

F. Prognosis

Prognosis is guarded in cases of severe forms. Brain hemorrhage is a dreadful complication as in the deficiencies of the other PC factors. Data on survival are scanty. It has surely improved with the widespread use of concentrates, but this beneficial effect has often been wiped out by the epidemic of blood transfusion-transmitted diseases. In recent years, this negative effect has faded and survival rates are increasing again. It has been estimated, in fact, that life expectancy for severe hemophilia patients without HIV infection is around 60 years, with no significant differences between hemophilia A and B (Darby *et al.*, 2007).

G. Therapy

Several variably purified concentrates, including recombinant FIX ones, are available for therapy (Kessler, 2005). Half-life is about 15 h, and safe hemostatic levels are around 30–40% of normal. This is based on the observation that carriers with a FIX level of about 40–50% of normal are asymptomatic.

Prothrombin complex concentrates were widely used in the past but have now been almost abandoned because of the high risk of viral contamination and of thrombotic complication due to the frequent presence in such concentrates of activated clotting factors. Recombinant FIX represents now the treatment of choice (Kessler, 2005; Roth *et al.*, 2001). Recombinant-activated FVII has also been used in critical conditions, but this is also associated with the high risk of thrombosis and should be discouraged (O'Connell *et al.*, 2006) unless inhibitors are present.

Antifibrinolytic drugs have been used as an adjunct to substitution therapy. Prophylaxis regimens with recombinant FIX are now used with 500 units twice a week for a 70–80 kg male. Development of inhibitors creates a severe therapeutic problem (Dimichele, 2007). Inhibitors develop in about 2% of hemophilia B patients. Since hemophilia B is at least six times less frequent than hemophilia A, the problem has, apparently, a smaller impact. However, this may be a fallacious impression due to the fact that the treatment of hemophilia B patients with inhibitors has so far less options than for patients with hemophilia A. Immune-tolerance results, for example, appear to be less favorable (Dimichele, 2007; Soto *et al.*, 2004). Activated PC concentrates or recombinant-activated FVII concentrate are the most widely used therapeutic tools (Dimichele, 2007). Immunoadsorption, cyclophosphamide, and cortisone have also been occasionally used with variable success.

Liver transplant has been carried out in a few instances, with apparent fair or good results, obtaining protracted normal or hemostatically satisfactory FIX levels (Johnston *et al.*, 1998; Kadry *et al.*, 1995; Kayler *et al.*, 2002; McCarthy *et al.*, 1996; Merion *et al.*, 1988; Sugawara *et al.*, 1997; Wilde *et al.*, 2002). The follow-up has been satisfactory in a few instances for 2–3 years. The procedure also appears to slow HIV progression. This approach is usually reserved for patients with hemophilia and liver failure due to hepatitis (Table 14.19) (Gordon *et al.*, 1998).

Table 14.19 Cases of liver transplantation in hemophilia B patients reported in the literature

References	Age of transplant	Type of hemophilia	Type of transplant	Type of hepatitis	HIV infection	Outcome	Comments
Merion *et al.* (1988)	60	Mild	Orthotopic	B	No	Good	
Kadry *et al.* (1995)	44	Severe	Orthotopic	B/C	No	Fair	Survival for 2 years
McCarthy *et al.* (1996)	40	Severe	Orthotopic	C	No	Poor	Death 20 days after transplant
Johnston *et al.* (1998)	60	Mild	Orthotopic	C	No	Good	
Kayler *et al.* (2002)	14	Severe	Orthotopic	n.r.	n.r.	Good	30 moths of follow-up
Sugawara *et al.* (2002)	41	Severe	Living donor	C	Yes	Good	Delayed HIV disease progression
Wilde *et al.* (2002)							Acute rejection.
Case 1	52	Severe	Orthotopic	C	Yes	Fair	Immunosuppressive therapy
Case 2	59	Moderate	Orthotopic	B	No	Poor	Acute renal failure

Gene therapy is also being investigated. Several vectors have been attempted, but the results obtained have been meager so far.

In a few studies it was employed, AAV mediated FIX gene transfer to skeletal muscle. Minimal and short-lived elevations of FIX were obtained in some of the patients investigated (Jiang *et al*., 2006; Manno *et al*., 2006). However, in one study enrolment was stopped because of increase in transaminases levels. Altogether, gene therapy for hemophilia B remains so far only a theoretical proposal (McCarthy *et al*., 1996; Ponder, 2006).

H. Combined defects of factor IX

FIX defect, hemophilia B, has been rarely described in association with Factor VIII deficiency (hemophilia A). The condition is due to the presence, in the same woman, of a double carrier state for hemophilia A and B. The number of cases reported is limited, around 20 cases (Robertson and Trueman, 1964; Shetty *et al*., 2001b).

The clinical picture is similar to that of severe hemophilia A or B.

The main features are a prolonged PTT partially but not completely corrected either by the addition of normal serum or by adsorbed normal plasma.

Another association that has obtained great attention is that with Factor VII (Battin *et al*., 1988; Bell and Alton, 1955; Constandoulakis, 1958). At least three conditions characterized by combined FIX and Factor VIII deficiency have been reported, namely, (1) the result of a casual association between hemophilia B with FVII deficiency. A few families have been described where the two defects segregate independently; (2) the result of laboratory artifact as seen especially in hemophilia Bm due to interference of the abnormal Factor IX in the Factor VII assay. This effect may be mild and go unnoticed when rapid tissue thromboplastins (rabbit or human brain) are used in the assay system, but may be evident when slow thromboplastins (ox-brain) are used; (3) the existence of a combined FIX and Factor VII defect due to a common cause. This last possibility remains to be proven. Needless to say that the combined defect of all PC factors is a separate clinical entity that will be treated separately.

V. Congenital Factor X Deficiency

Since Factor X (FX) plays an important role in blood coagulation, it is not surprising that the discovering of a congenital deficiency of this factor represented a major event in blood coagulation.

The main problem concerned the differences from Factor VII deficiency, which, being less rare, was described before FX, namely, in 1951.

Such separation between FVII and FX deficiency with the description of the first cases of isolated congenital FX deficiency was accomplished in 1956 and 1957 with the description of the three kindreds (Power in England, Stuart in USA, and another in Switzerland). Several cases have been added afterward.

Note that the classical FX deficiency in the Stuart family had been described before as a case of hypoconvertinemia (Graham, 2003). The same was true for two Factor X Friuli patients (Girolami *et al.*, 1970), who were first described as cases of FVII deficiency and subsequently, after the description of FX deficiency, the patients were restudied and described as cases of FX deficiency (Chevallier *et al.*, 1955, 1959). This represented an improvement but the final diagnosis of an abnormal FX arrived later when anti-FX antiserum became available (Girolami *et al.*, 1970). Several reviews have appeared during the past decade (Cooper *et al.*, 1997b; James, 1994; Perry, 1997; Uprichard and Perry, 2002). A great heterogeneity of the defect has been demonstrated (Denson *et al.*, 1970; Fair and Edgington, 1985).

A. Classification and prevalence

Factor X defects should be classified as: (1) "true" deficiency, (2) FX abnormalities, and (3) combined defects. This classification is based on both activity and antigen assays. Families investigated only on activity assays cannot be adequately evaluated even if molecular biology data are available (Herrmann *et al.*, 2006).

Similarly to what already seen with other factors of the PC, the patients with "true" deficiency show a parallel decrease in FX activity and antigen. This form is also called Type I.

Patients with abnormal FX show low activity and slightly decreased or ever-normal antigen, the so-called Type II. The cases with normal FX antigen are usually referred to as CRMpos, those with intermediate levels as CRMred, whereas the cases with "true" deficiency are defined as CRMneg. Approximately 1/3 of patients are cases of true deficiency; in remaining 2/3, FX antigen is present, either in normal or in reduced quantity. On the basis of cases so far described, a simple classification has been proposed (Girolami, 1986): true deficiency (Stuart defect) (Graham *et al.*, 1957; Hougie *et al.*, 1957), cases with low activity and normal antigen (Prower defect) (Blanchette *et al.*, 1991; Cooper *et al.*, 1997b; Denson *et al.*, 1970), forms with normal RVV (Friuli defect) (Girolami, 1971; Girolami *et al.*, 1971; James *et al.*, 1991; Kim *et al.*, 1996) and forms with defect only in the extrinsic system (Padua and similar defects) (Al-Hilali *et al.*, 2007; Bertina *et al.*, 1981; Nora *et al.*, 1985; Parkin *et al.*, 1974), and forms with a defect only in the intrinsic system (FX Melbourne) (Parkin *et al.*, 1974) (Table 14.20).

These five types represent the index patients who have a distinct laboratory pattern and variable clinical picture. The majority of cases described in

Table 14.20 Tentative classification of Factor X deficiency and paradigmatic defects

Condition	Factor X assay					
	Tissue thromboplastins (extrinsic)	RVV	Cephalin (intrinsic)	Chromogenic	Immunological assay	Bleeding manifestations
Classic FX Def. (Mr. Stuart)[a]	Low	Low	Low	Low	Low	Severe
Miss Prower Def.[b]	Low	Low	Low	?	Normal	Severe
FX Friuli Def.[c]	Low	Normal	Low	Low	Normal	Moderate
FX Melbourne	Normal	Normal	Low	?	Normal	Moderate
FX Padua[d]	Low	Normal	Normal	Low	Normal	None

[a,b] Several similar cases described.
[c] No similar cases reported.
[d] A few additional similar cases reported (FX San Antonio, FX Riadh).

recent years by molecular biology techniques fit either the Stuart or the Prower genotype. In other words, they are either cases of "true" deficiency with no CRM or cases with variable levels of CRM on their plasma (Bernardi *et al.*, 1989; Bezeaud *et al.*, 1995; Fair *et al.*, 1989; Forberg *et al.*, 2000; Girolami *et al.*, 1976; Hayashi *et al.*, 1998; Herrmann *et al.*, 2006; Huisse *et al.*, 1985; Kim *et al.*, 1996; Marchetti *et al.*, 1995; Miyata *et al.*, 1998a; Peyvandi *et al.*, 2002; Pina-Cabral and Justica, 1973; Simioni *et al.*, 2001; Watzke *et al.*, 1990; Zama *et al.*, 1999).

Combined defects refer to cases of Factor X defect in association with other clotting defects.

The estimated prevalence of FX deficiency is about 1/1,500,000 or 2,000,000, therefore the condition to be considered as one of the rarest coagulation disorders.

B. Biochemistry and function

FX is a vitamin K-dependent glycoprotein, synthesized in the liver as a single chain. It circulates as a two chain at a concentration of about 1 mg%.

Its structure is similar to that of other PC clotting factors. The circulating double-chain form consists of a light chain of 139 amino acids and a heavy chain of 346 residues bound by disulfide and other bonds (Hertzberg, 1994; James, 1994).

The light chain contains three structural domains, whereas the heavy chain contains the serine protease domain that forms the catalytic triad of the active site.

The activation of FX occurs by the cleavage of the Arg194–Ile195 bond of the heavy chain. Such activation occurs physiologically either by the intrinsic system (FIXa and Factor VIIIa, phospholipids, Ca^{2+} ions) or by the extrinsic system (TF, FVIIa, Ca^{2+}, and phospholipid membrane). Nonphysiological activation is represented by RVV because of the presence in this venom of a metalloproteinase. The main substrate of FXa is prothrombin; other minor substrates are Factors V and FVII.

The FXa activity is regulated by natural occurring inhibitors, namely, antithrombin, which is a direct inhibitor. An indirect inhibitor may be considered the TFPI. This inhibitor forms a quaternary complex with TF, FVIIa, and FXa devoid of any activity. Recently, an inhibitory effect on FXa has been proposed also for the protein Z-dependent protease inhibitor (Broze, 2006). However, its significance remains to be clarified.

C. Hereditary transmission and molecular genetics

The hereditary transmission of FX deficiency is autosomal recessive. People who are homozygous for true deficiency have very low FX levels and equivalently depressed FX antigen.

Those who are heterozygous have FX levels of about 50% of normal levels both as activity and as antigen.

In homozygous FX abnormalities, FX activity level is variably depressed, usually around 5–15% of normal, while normal or near normal antigen is present. Heterozygotes, in this case, have activity of about 50–60% of normal and normal FX antigen levels. In compound heterozygotes (hipo-dys or dys-dys), FX levels are variable. For example in the association of a true deficiency with an abnormality (hypo-dys form), FX level is usually around 50–60% of normal both as activity and as antigen (Fig. 14.7).

In compound heterozygous, the defects segregate independently in the family so that some family members may have only one of the two defects.

The human FX gene is 22-kb long and is located at 13q34-ter, <3 kb from FVII. The sequence of human FX gene is highly homologous to other vitamin K-dependent coagulation factors such as FVII, FIX, and Protein C, suggesting its origin from a common ancestral gene. The coding region of FX is divided into eight exons, each encoding for a particular domain within the protein. Exon I encodes the signal peptide, Exon II the propeptide and the Gla domain, and Exon III a short-linking segment of aromatic amino acids. Exons IV and V encode the two EGF domains; Exon VI encodes the activation peptide at the N-terminal of the heavy chain; and Exons VII and VIII encode the serine protease domain with the catalytic triad His236, Asp228, and Ser379 (Table 14.21).

Although the clinical presentation of FX deficiency correlates fairly well with laboratory phenotype, genetic analysis may sometimes provide a further explanation for the FX activity and antigen levels measured in the laboratory. As in the other congenital coagulation disorders, FX deficiency usually arises from missense mutations. Amino acid substitutions can significantly change any number of steps in protein formation. Mutations severely impairing the protein structure are frequently associated with a reduction in antigen and consequently in the activity level. In these cases, abnormal protein folding may be responsible for an impairment of protein secretion and/or a reduction in protein half-life. Missense mutations responsible for minor structural changes are often associated with an impairment in FX activity in the setting of a normal antigen level. Large gene deletions have been reported and are invariably associated with a symptomatic pattern when found in compound heterozygosis with a missense mutation impairing the FX protein structure. Very few symptomatic individuals with total or severe deficiency in the components of prothrombinase have been identified to date (Cooper *et al.*, 1997b; Millar *et al.*, 2000; Oldenburg *et al.*, 2007). It is likely that genetic mutations that cause total FX deficiency in humans may often be incompatible with life. Missense mutations have been described affecting the different FX domains. Mutations in the pre-prosequence and prosequence, as FX Nice and FX Santo Domingo, have been shown by molecular modeling and expression studies to be responsible

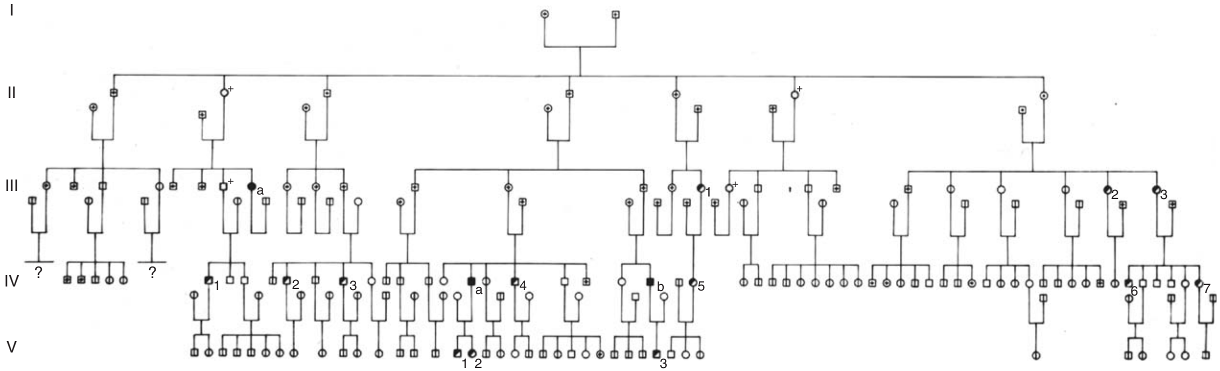

Figure 14.7 The FX Friuli index kindred. The hereditary transmission is autosomal recessive. Consanguinity is evident. Homozygotes are IIIa (index patient) and her two relatives (IVa and Vb). Numerous heterozygotes are present in the family tree. Classical FX deficiency has the same hereditary pattern. Symbols have the same meaning as in Fig. 14.1.

Table 14.21 Main features of some representative patients with variably severe "true" FX deficiency. Eponyms have been disregarded

References	FX activity				FX Antigen	Bleeding tendency	Genotype	Mutation
	Intrinsic	Extrinsic	RVV	Chromogen (S2222)				
Graham *et al.* (1957) and Cooper *et al.* (1997b)	<1	<1	Low	n.r.	Traces	Severe	Hom.	Stuart pedigree
Watzke *et al.* (1991)	1	<1	n.r.	n.r.	<5	Severe	Hom.	Val 289 Met Gly -20 Arg
Miyata *et al.* (1998a)	n.r.	3	n.r.	n.r.	<10	Moderate	Hom.	Arg 306 Cys
Miyata *et al.* (1998b)	n.r.	4	n.r.	n.r.	7	Moderate	Comp. Het.	Met 40 Val + Pro 204 Ser
Hayashi *et al.* (1998)	n.r	25	4.5	n.r.	<5%	Moderate	Hom.	Triplet deletion intron D
Forberg *et al.* (2000)	30	1	18	n.r.	20	Moderate	Double Hom.	Gla 129 Lys + Glu 102 Lys
Peyvandi *et al.* (2002)								
H	<1	2	1.5	9	4	Severe	Hom.	Gly 94 Arg
J	<1	<1	<1	3	10	Severe	Hom.	Gly 222 Arg
Vianello *et al.* (2003)	4	4.7	3.9	2.5	5	Moderate	Hom.	Cys 350 Arg

List is obviously incomplete.
Letters are those proposed by authors to define their families. RVV, Roussell viper venom; n.r., not reported; Hom., Homozygote; Comp. Het., Compound heterozygote; Het., Heterozygote.

for secretion and cleavage problems explaining the Type I phenotype (Miyata *et al.*, 1998b; Watzke *et al.*, 1991).

A missense mutation affecting the calcium-binding region of the Gla domain characterizes the low activity and normal antigen level in the FX St. Louis (Rudolph *et al.*, 1996). Mutations affecting the correct formation of disulfide bridges in EGF-1 and EGF-2 domain have been reported, affecting both the function and the antigen level of FX-associated mutations. Missense mutations affecting the release of the activation peptide, such as FX Wenatchee I and II, have been shown to be associated with low-catalytic domain. This variant consists of a Pro343Ser substitution within the heavy chain of FX (Kim *et al.*, 1995). Homozygous FX Friuli patients present with a normal antigen level and a normal or near normal activity by means of RVV assay, but a severely reduced (4–9%) function in the intrinsic and extrinsic pathways. Because of the formation of a new hydrogen bond, the tertiary structure of the catalytic domain has been shown to be affected. A good genotype–phenotype correlation has been demonstrated in an Italian family with FX deficiency, named FX S. Giovanni Rotondo (Simioni *et al.*, 2001). In this variant, one deletion in the region encoding for the activation peptide was described in compound heterozygosis with a missense mutation in the catalytic domain, respectively, responsible for the lack of synthesis and/or secretion of FX and for a dysfunctional FX (Table 14.22). The loss of a disulfide bond in the catalytic domain has been shown to be associated with a Type I phenotype in FX Padua 3. These few examples of a consistent genotype–phenotype relationship further suggest that the laboratory phenotype is largely a function of the FX gene lesion segregating in the family. Some variability, even in the presence of identical pathological FX genotypes, may be present, due to variation between laboratory assays, the effect of polymorphisms, and possibly also the influence of variation in nonallelic loci. In other cases, the explanation for these discrepancies remains unknown. In this regard, it is worth remembering that the FX variants with a defect only or predominantly in the extrinsic system (FX Padua, FX Riadh, and FX San Antonio) show different mutations, namely, Arg251Trp, Glu51Lys, or Arg326Lys + 272 Stop, respectively (Al-Hilali *et al.*, 2007; Girolami *et al.*, 1985b, 2004; Reddy *et al.*, 1989). Another example of the lack of a constant relation between genotype and phenotype. FX Roma (De Stefano *et al.*, 1988; Odom *et al.*, 1994) shows a pattern similar to that of FX Friuli (Girolami *et al.*, 1970; James *et al.*, 1991), namely, prolonged PTT and PT and normal RVV and antigen. This could suggest that the area of Exon 8 between codon 318 (Roma) and 343 (Friuli) is involved in the activation via the RVV. However, this is not absolute, since the Ser334Pro mutation (FX Marseille) shows a prolonged RVV clotting time (Bezeaud *et al.*, 1995). Another discrepancy that makes it so far very difficult to draw sure genotype–phenotype conclusions.

Table 14.22 Main features of the 5 FX abnormalities seen in Padua during the past 37 years. Cases with true deficiencies have been excluded

Abnormality	FX activity				Antigen	Bleeding	Genotype	Mutation and comments
	Extrinsic	Intrinsic	RVV	Chromogen (S2222)				
FX Friuli	8	6	90	2	100	Moderate	Hom. and Het.	Pro343Ser
FX Padua 1	76	90	88	20	100	Mild	Hom. and Het.	Arg251Trp
FX Padua 2	52	60	56	40	96	Moderate	Het.	Molecular biology studies not available.
FX San Giovanni Rotondo	9	24	43	22	72	Moderate	Comp. Het.	Lys408Pro microdeletion C556
FX Padua 3	8.2	19	12	10	40	No	Comp. Het.	Gly 380 Arg Ser 334 Pro[a]

[a] First described in FX Marseille. RVV, Roussell viper venom; Hom., homozygote; Comp. Het., compound heterozygote; Het., heterozygote.

Expression studies, which allow to obtain recombinant forms of the FX been investigated, are essential to confirm that the mutation observed is indeed responsible for the defect. These studies are particularly indicated when dealing with abnormalities but, unfortunately, have been carried out only in a few cases (Forberg *et al.*, 2000; Kim *et al.*, 1996; Racchi *et al.*, 1993).

Molecular studies have recently allowed the construction of putative models that may help in understanding the structure–function relation.

D. Clinical picture

Among patients with rare coagulation defects, FX deficiency may be one of the most severe. The onset of hemorragic manifestations in marked deficiency is typically seen in infancy. In less severe deficiency, patients may start experiencing bleeding at any age. The most frequent sites of bleeding are the mucosal, soft tissue, and central nervous system. Easy bruising is also frequent. Among mucosal bleeding, epistaxis is the most frequent, followed by menorrhagia and gastrointestinal episodes. Soft tissues and joint bleeding such as muscle hematomas and hemarthroses may mimic classic hemophilia but are rare. Central nervous system hemorrhages represent life-threatening manifestations of severe FX deficiency. This is a relatively frequent cause of death as in other defects of the PC. It seems almost a specific symptom of the condition. Another severe bleeding complication is represented by hemoperitoneum associated with ovulation (Dafopoulos *et al.*, 2003).

There is no specific hemorrhagic pattern, and no sure diagnosis can surely be established on clinical grounds. Altogether, the picture is fairly similar to that of FII or FVII deficiency. Individuals with mild FX deficiencies may experience bleeding only after trauma or surgery.

Heterozygotes with FX levels around 50% of normal are usually asymptomatic but may present sometimes, contrary to what seen in heterozygotes for FVII deficiency, with easy bruising and bleed during surgery, traumas, and delivers.

E. Diagnosis

Clinical signs and symptoms are not specific. They may vary in intensity according to the plasma levels. Diagnosis of FX deficiency can be reached only by specific laboratory investigation. The main features are: prolonged PTT PT with normal TT. RVV clotting time is also prolonged in the majority of cases, but it may be normal in some variants (FX Friuli). FX assays have to be carried out using at least three activation systems, namely, intrinsic, extrinsic, and RVV (Bachmann *et al.*, 1958; Girolami *et al.*, 1970). This is necessary because variants in which only one or two of these assays were abnormal have described. FX Padua, for example, yielded a defective pattern only in the extrinsic system (Girolami *et al.*, 1985b, 2004). A few other cases with similar clotting features have been described (FX San

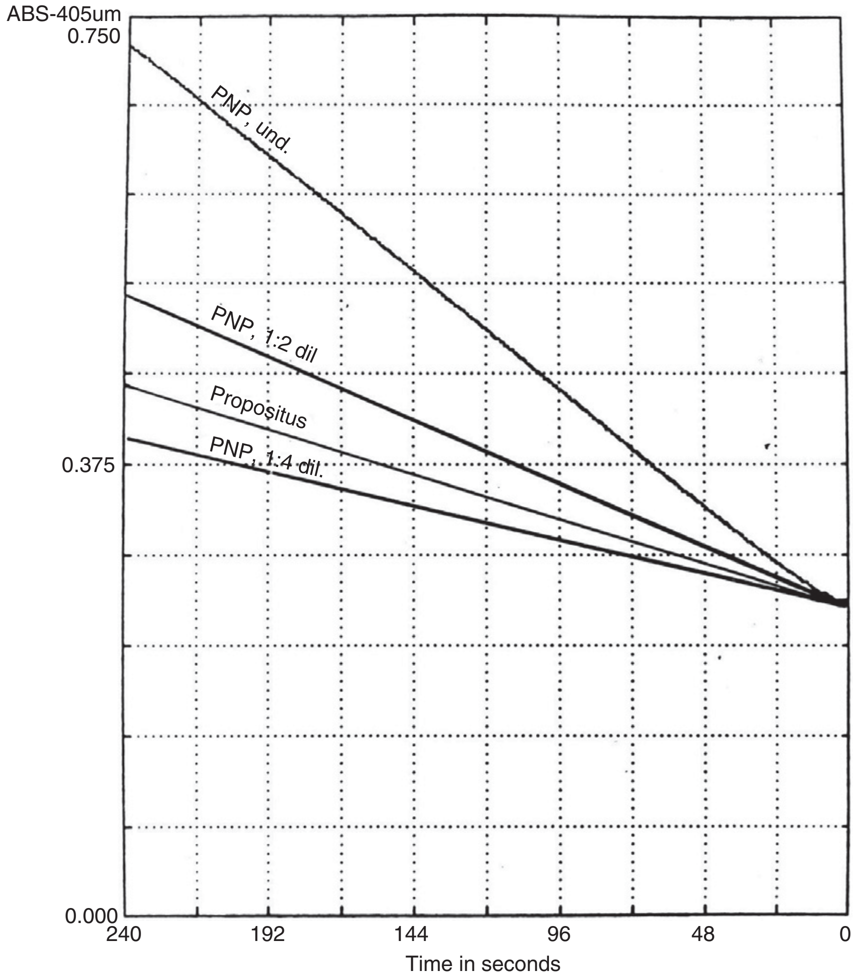

Figure 14.8 Chromogenic assay (S222). The propositus (a case of FX Padua 2) showed a level of about 38% of normal. Antigen level in this patient was 100% of normal.

Antonio and FX Riadh) (Al-Hilali *et al.*, 2007; Reddy *et al.*, 1989). Chromogenic assays may also be useful since, sometimes, they may yield different levels of activity as compared with the clotting counterparts (Girolami *et al.*, 1980a) (Fig. 14.8).

These peculiar behaviors complicate the differential diagnosis in comparison with FVII deficiency. This is more so when one deals with cases of multiple factors deficiency, for example, combined Factor VII and Factor X defect (Table 14.20).

An immunological assay (usually, electroimmunoassay, radial immunodiffusione, ELISA, or laser nephelometry) is also indicated to find out if there is a discrepancy between activity and antigen level. In true deficiency, there is no discrepancy (CRMneg). In other forms, there may be an excess without reaching normal levels of antigen and activity (CRMred) (Girolami *et al.*, 1971; Mancini *et al.*, 1965; Vianello *et al.*, 2001, 2003).

When FX activity is low and FX antigen is normal or near normal, we have the so-called CRMpos. Genetic analysis does not add substantial information about diagnosis and clinical picture but is important for structure–function studies.

Molecular diagnosis of congenital FX deficiency relies on the detection of gene mutation and on the possible genotype–phenotype correlation. This can be achieved by amplifying the entire coding of the gene by PCR and then proceeding to sequence the PCR products using the appropriate primers (Fig. 14.9). Even with the introduction of automated sequencing system, some intermediate steps can be useful for scanning PCR products with the aim to identify the gene fragment potentially mutated and then confirm the mutation by sequencing (Fig. 14.10). The scanning techniques most commonly used for PCR products are single strand conformation polymorphism (SSCP), enzymatic or chemical cleavage of mismatched base pairs, and differential unfolding of homoduplex and heteroduplex by denaturing gradient gel electrophoresis (DGGE). Recently, CSGE has been successfully used as a simple, inexpensive, and highly reliable method for localizing FX mutations (Fig. 14.11). The combination of PCR CSGE with sequencing could be useful in family studies where, after complete characterization of the mutation in a family member by sequencing the entire FX gene, the other relatives can be screened by means of CSGE only (Vianello *et al.*, 2002). Although expression studies are invaluable to detect subtle functional differences in the cases where the mutations are antigen positive (i.e., normal antigen levels), they are less useful where the mutations are antigen negative (as is reflected in the low *in vivo* plasma levels), as the mutant proteins are likely to be unstable or may not be expressed. With the availability of FX crystal structures, sequence alignments and molecular modeling are effective methods in identifying structural perturbations in active sites (Vianello *et al.*, 2003). Molecular modeling in combination with energy refinements can be invaluable tools in the assessment of cases in which protein misfolding is the most likely outcome of a given mutation (Fig. 14.12). Molecular biology techniques have allowed a prenatal diagnosis in a few cases (Ingerslev *et al.*, 2007). In the context of a family with homozygotes, heterozygotes can easily be detected by repeated FX assays. There is usually little overlap, if any, between low normals and heterozygotes (Fig. 14.13) (Girolami *et al.*, 1974c). When dealing with sporadic cases, diagnosis may be more difficult.

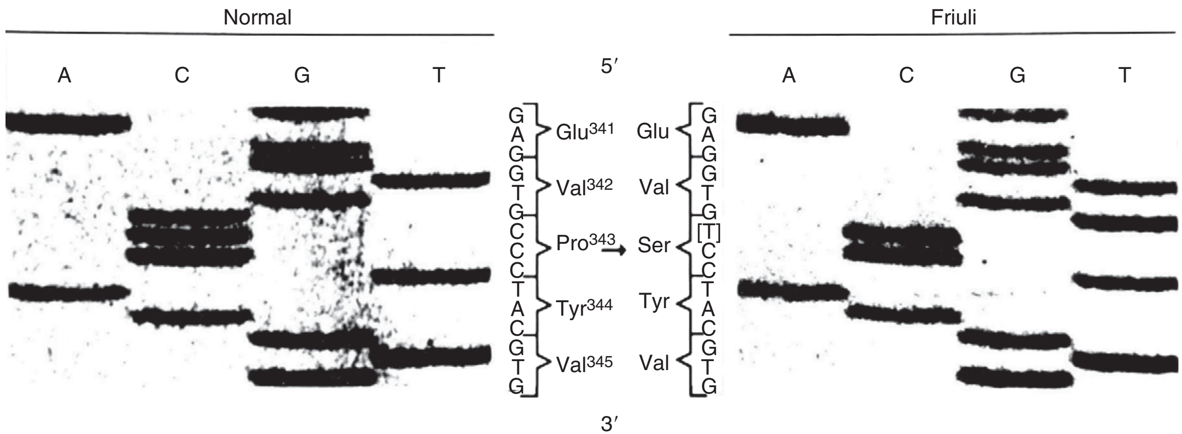

Figure 14.9 Autoradiographs of Exon VIII of normal FX and FX Friuli genes. The mutation results in coding for serine (TCC) instead of praline (CCC) at position 343 in the FX Friuli molecule.

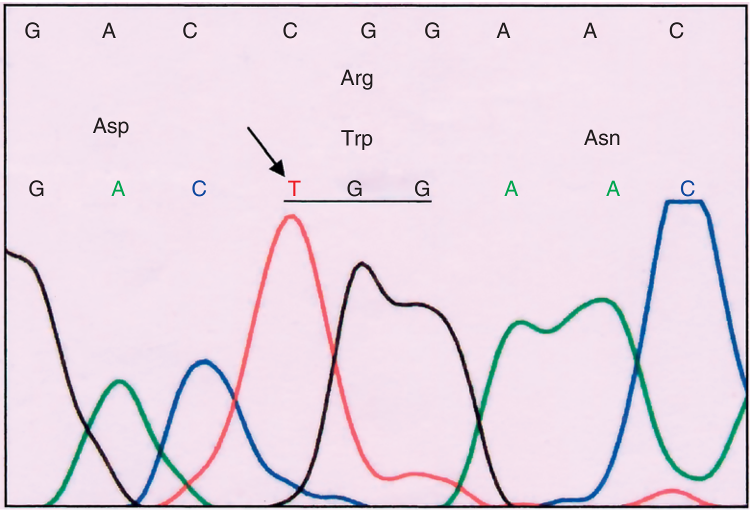

Figure 14.10 Sense DNA sequence of Exon VIII of FX Padua 3 gene resulting in the replacement of Arg with Trp at position 251.

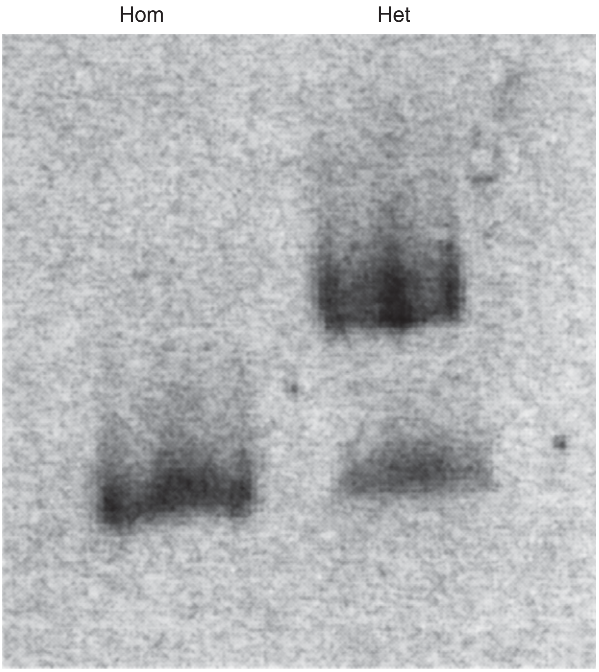

Figure 14.11 Conformation sensitive gel electrophoresis (CSGE) of PCR amplified Exon 8 in a Factor X abnormality (Arg251Trp, or Padua). This is a useful screening procedure for clotting factor abnormalities with normal or near normal antigen in their plasmas. Hom., homoduplex; Het., heteroduplex.

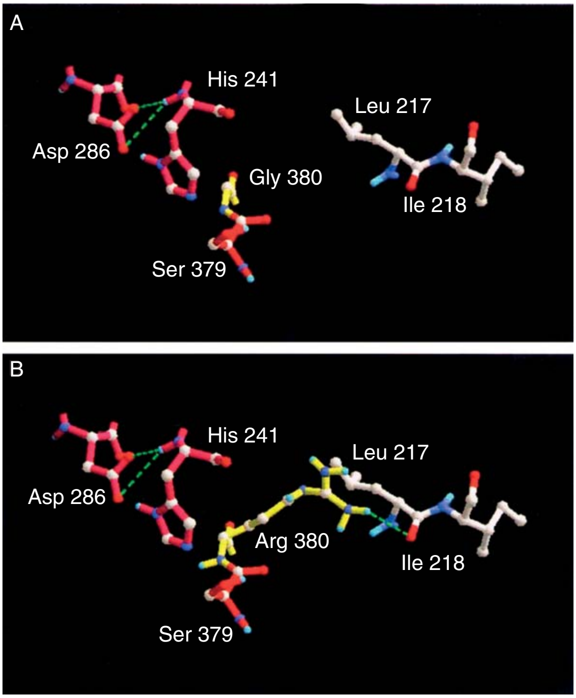

Figure 14.12 Putative tridimensional structure of normal FX and of FX Padua 3 characterized by the Gly380Arg mutation. Catalytic residues are shown in red; site of mutation in yellow; broken lines indicate hydrogen bonds.

F. Prognosis

Prognosis varies with the severity of defects. The limited number of cases described has not allowed the gathering of sure information on survival. Fatal brain hemorrhage is an important cause of death in early age. Levels of about 10% of normal seem sufficient to assure a decent life, providing no surgery, trauma, pregnancy, or invasive diagnostic procedures are needed. These levels may even afford some antithrombolitic protection. In fact, one of the patients of the FX Friuli is alive and fairly well at the age of 103 (Girolami *et al.*, 2005). This is the only study of prognosis available, but it refers only to a specific abnormality (FX Friuli) in which homozygotes have FX activity levels around 6–8% of normal. It is also worth noting that in another of these patients who died at the age of 72 for a carcinoma of the pancreas, arteries were found, at autopsy, smooth for the age (Girolami *et al.*, 1975a).

G. Therapy

Plasma or PC concentrates are the available tools. No FX concentrate is available. PCC are usually titrated in FIX activity, and therefore their content in FX may vary. The relatively long half-life allows spaced

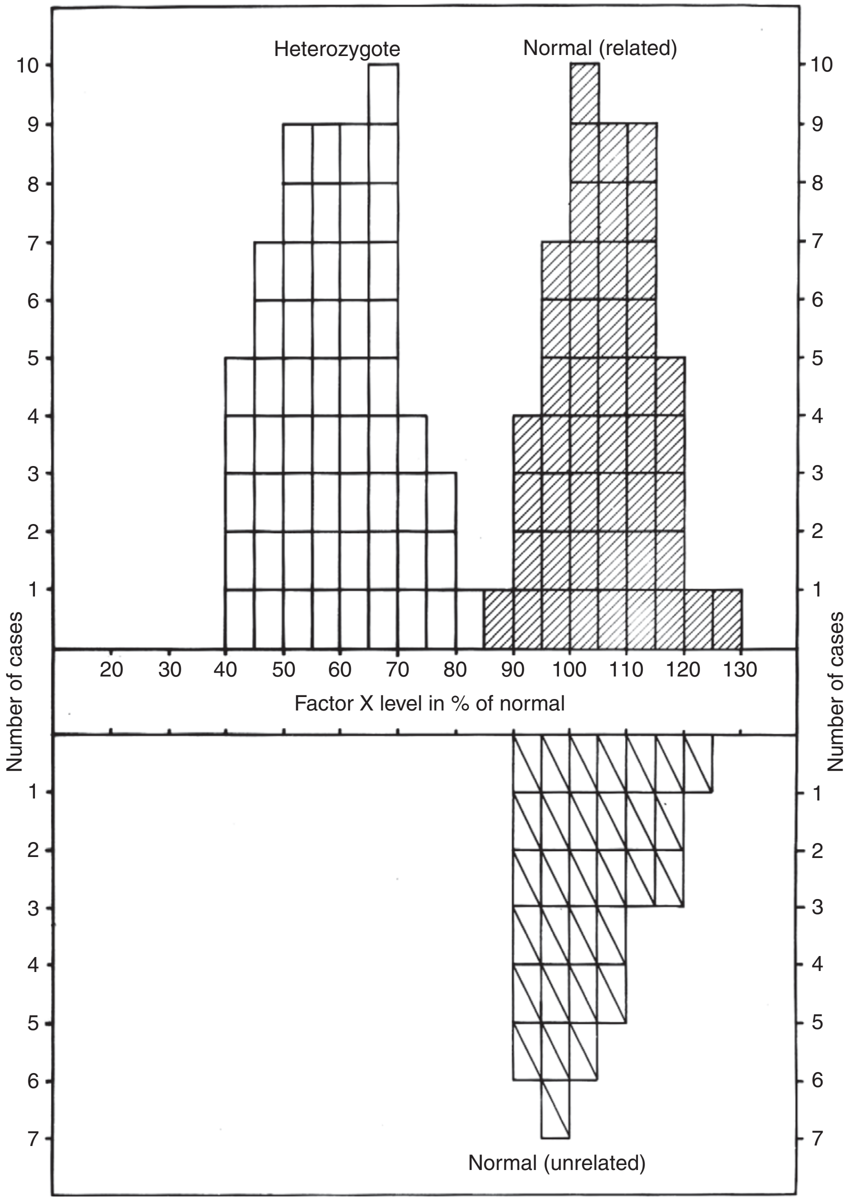

Figure 14.13 Factor X (FX) levels as determined by the extrinsic system in 57 heterozygotes for FX Friuli disorder and in 77 normal subjects. The normal subjects were both related and unrelated to the homozygote patients. The figure entered is the mean of at least two determinations carried out on different occasions. The overlay between affected and normal appears minimal, if any.

administrations. Safe hemostatic levels are at least around 40–50% of normal in case of surgery, trauma, and delivery. This is based on the observation that heterozygotes who have similar or slightly higher levels, only rarely bleed spontaneously (Figs. 14.14 and 14.15).

However, even heterozygotes may occasionally need transfusional support in cases of major surgical procedures.

Spontaneous bleeding in homozygotes or other patients (compound heterozygotes) with FX levels of around 10% is uncommon. Prophylaxis has been carried out in a few severe cases with good results (Auerswald, 2006). A satisfactory reduction of bleeding manifestations can be achieved, especially in children, with the administration of PC concentrates containing also FX with the weekly or biweekly infusion of 10–20 units/kg body

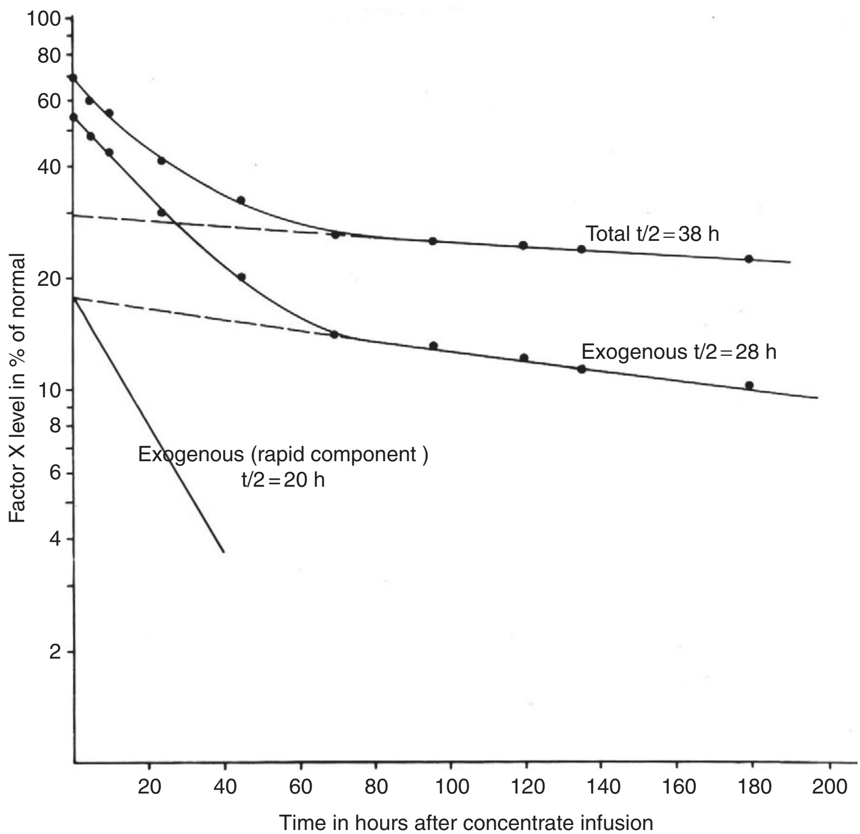

Figure 14.14 Factor X (FX) survival time after infusion of a prothrombin complex concentrate (2,000 units as titrated for FIX) in a patient with FX Friuli disorder (basal FX activity 5% of normal) before cholecystectomy. The survival time of exogenous compound is about 20 h. Surgical bleeding has been minimal.

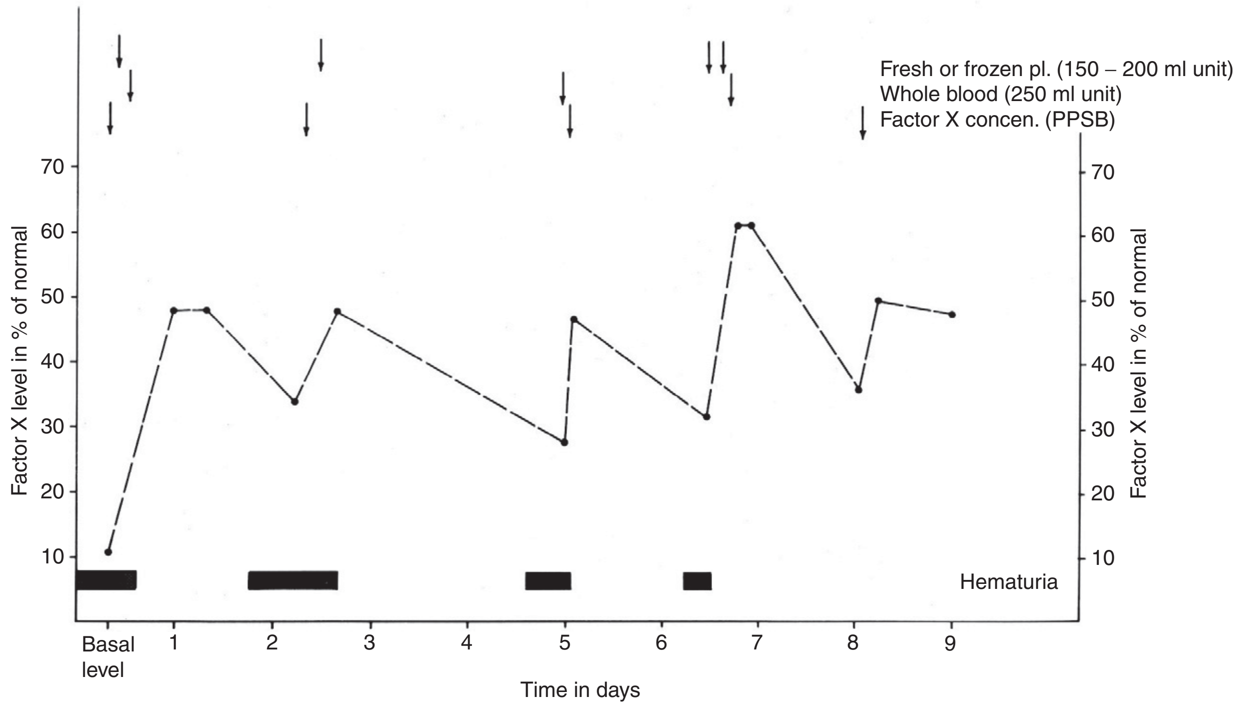

Figure 14.15 Variation of Factor X (FX) activity levels in a patient with FX deficiency being treated for persistent hematuria. The hemostatically safe levels seem to be around 45–50% of normal. In fact, hematuria reappeared whenever FX activity was below 40%. Arrows at top of the figure indicate time of transfusional therapy, as indicated.

weight (Auerswald, 2006). Oral contraceptives have been reported to be effective in reducing blood loss and in improving the anemia (Girolami *et al.*, 2006; Payne *et al.*, 2007). Pregnancies have been carried to term with adeguate prophylaxis (Girolami *et al.*, 2006; Romagnolo *et al.*, 2004).

Antifibrinolytic agents have been used as adjunct measures. Orthotopic liver transplant has been carried out with success in at least one patient (McCarthy *et al.*, 1996). No attempts at genetic therapy have ever been reported.

H. Combined defects

FX deficiency has been rarely described in association with other clotting factors, namely, FVII, hemophilia A, and Factor XII deficiencies. The rarity of the disease may fully justify the scanty of reports.

The most important association seems to be that with FVII, as we have stated dealing with this factor. Two types of this association seem to exist, the first type due to the casual association of the two clotting factors that segregate independently in a given family. The second type seems to exist due to alterations of the long arm of chromosome 13 where both FX and FVII genes are located (13q34). The two genes are very close, and therefore deletions or other abnormalities may involve the two genes (Kasai *et al.*, 1989). Several cases have been described and patients present, usually, additional alterations, besides clotting defect (mental retardation, cardiac abnormalities, etc.). In this form, the hereditary pattern seems autosomal dominant.

Association with other clotting factors is rare and due to the casual combination of two, independently segregating defects.

VI. Combined Congenital Deficiency of FII, FVII, FIX, FX

The congenital combined deficiency of all the PC factors including Protein C and Protein S is a rare but fascinating disease. It is often called VKCFD (vitamin K complex factor deficiency), and it has received considerable attention in recent years (Zhang and Ginsburg, 2006).

Only 28 cases belonging to 23 families can be gathered from the literature plus one so far unpublished from our files for a total of 29 cases.

The defect may therefore be considered as the rarest coagulation disorder.

The defect is due to mutation either in the γ-carboxylase or in the epoxide reductase systems. These enzymes play a critical role in the vitamin K metabolism or cycle.

A. Classification and prevalence

The defect is usually subdivided in two groups, namely, VKCFD Type I due to defective γ-carboxylase activity and VKCFD Type II. The Type II VKCFD is the result of a functional deficiency of vitamin K 2,3-epoxide reductase. This is the same enzyme that is involved in coumarin drugs therapy.

The first case was seen by Newcomb *et al.* (1956). Even though it was described before the separation of Factor X from Factor VII, a careful evaluation of the paper allows the conclusion that the propositus was indeed a case of congenital combined PC deficiency. The second case was the one presented by McMillan and Roberts (1966) and restudied by Chung *et al.* (1979).

At least 27 additional cases have been studied afterward (Table 14.23).

B. Biochemistry and function

The biochemical features of these factors have been dealt with while presenting the single factor deficiencies. They are all glycoproteins with important structural similarities.

γ-Glutaryl carboxylation of these proteins is needed for their activation. Such carboxylations allow for these proteins to achieve calcium-dependent structural changes necessary for binding to membrane phospholipids. The γ-carboxylation process also involves noncoagulation-related proteins, for example osteocalcin, and this may explain the occasional association in these patients with skeletal or other abnormalities.

Vitamin K is an important cofactor of the γ-carboxylation process (Oldenburg *et al.*, 2007). Vitamin K epoxide reductase (VKOR) is required to reconstitue the vitamin K hydroquinone form (Fig. 14.16).

Note that these hypocarboxylated or acarboxylated forms are present in both types of the condition. This is due to the fact that the defect in the epoxide reductase interferes with the carboxylation process by reducing the availability of reduced vitamin K (vitamin K hydroquinone).

C. Hereditary transmission and molecular biology

The hereditary transmission seems autosomal recessive, but no definite pattern has been established. Both homozygous and heterozygous have been occasionally described, but distinction is often not clear since parents of propositi show often a normal coagulation pattern. Both sexes are equally affected. There seems to be no sure ethnic preference, since the defect has been reported in several countries (Bhattacharyya *et al.*, 2005; Brenner *et al.*, 1998; Chung *et al.*, 1979; Fischer and Zweymuller, 1966; Ghosh *et al.*, 1996; Goldsmith *et al.*, 1982; Johnson *et al.*, 1980; Le Corvaisier-Pieto *et al.*,

Table 14.23 Cases of combined prothrombin complex deficiency described in the literature

References	Age (years), sex	Consanguinity	Degree of defect	Correction by vitamin K	Molecular biology	Associated abnormality
Newcomb *et al.* (1956)	4, F	n.r.	Severe	Partial	n.r.	No
McMillan and Roberts (1966) and Chung *et al.* (1979)	0.4, F 15	No	Severe	Partial	n.r	No
Fischer and Zweymuller (1966)	21, F	n.r.	Severe	Partial	n.r.	n.r.
Girolami *et al.* (1975)[a]	27, F	No	Moderate	Partial	n.r.	No
Mickelson and White (1979)	0.1,	n.r.	Severe	Partial	n.r.	No
Johnson *et al.* (1980)	2, M	No	Mild	No	n.r.	No
Goldsmith *et al.* (1982)	20, F	No	Mild	Yes	n.r.	No
Vicente *et al.* (1984)	1.2, F	No	Severe	No	n.r.	No
Ekelund *et al.* (1986)	14, M	n.r.	Severe	No	n.r.	No
Pauli *et al.* (1987)	7, M	No	Moderate	Partial	VKOR	Skeletal
Leonard (1988)	n.r.	n.r.	n.r.	Yes	n.r.	Yes
Le Corvaisier-Pieto *et al.* (1996)	40, M	No	Mild	n.r.	n.r.	Pseudoxanthoma elasticum
Boneh and Bar-Ziv (1996)	0.4, F	Yes	Moderate	Yes	n.r.	Skeletal
Ghosh *et al.* (1996)	4, M	No	Moderate	Partial	n.r.	Mental retardation
Brenner *et al.* (1998)						
Case 1	14	No (same family)	Moderate	Partial	Carboxylase	Skeletal

(continued)

Table 14.23 (*continued*)

References	Age (years), sex	Consanguinity	Degree of defect	Correction by vitamin K	Molecular biology	Associated abnormality
Case 2	7		Moderate	Partial	Id	No
Case 3	5		Moderate	Partial	Id	No
Case 4	4		Moderate	Partial	Id	No
Oldenburg *et al.* (2000)						
Case1	n.r.	Yes	Mild	Yes	VKOR	No
Case2	n.r.	No	Mild	Yes	VKOR	No
Spronk *et al.* (2000)	0.1, M	Yes	Moderate	Partial	Carboxylase	No
Mousallem *et al.* (2001)	0.2, n.r.	Yes	Severe	Partial	Carboxylase	No
McMahon and James (2001)	34, F	No	Mild	Partial	n.r.	No
Thomas and Stirling, 2003						
Case1	3	Yes	Mild	Partial	Carboxylase (deletion)	No
Case2	n.r.	Yes	Mild	Partial	Carboxylase	No
Puetz *et al.* (2004)	0.1, M	n.r.	Moderate	Partial	n.r.	Nervous system
Rost *et al.* (2004)	1, M	No	Mild	Partial	Carboxylase	Arteriosus ductus
Bhattacharyya *et al.* (2005)	n.r., M	Yes	Moderate	Partial	n.r.	No
Rost *et al.* (2006)	38, F	n.r.	Mild	Partial	Carboxylase	n.r.

[a] Unpublished observation.
n.r., not reported or not available; VKOR, Vit K epoxide reductase.

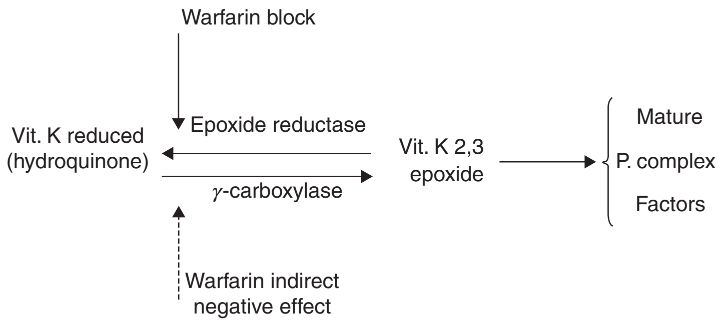

Figure 14.16 Schematic representation of vitamin K role in the formation of mature prothrombin complex factors. Vitamin K is needed for γ-carboxylase activity which is essential for the formation of active clotting proteins. Vitamin K epoxide is formed during the carboxylation process and has to be regenerated to active form (hydroquinone) by the vitamin K epoxide reductase (VKOR) system. This enzyme is blocked by Warfarin. Broken arrow indicates an indirect negative effect on γ-carboxylase activity due to decreased amounts of hydroquinone availability during Warfarin administration.

1996; Leonard, 1988; McMahon and James, 2001; McMillan and Roberts, 1966; Mickelson and White, 1979; Mousallem *et al.*, 2001; Newcomb *et al.*, 1956; Pauli *et al.*, 1987; Perry, 2002; Reitsma *et al.*, 1988; Rost *et al.*, 2004, 2006; Spronk *et al.*, 2000; Thomas and Stirling, 2003; Vicente *et al.*, 1984). Consanguinity is present in about 50% of families in which this aspect was looked for and reported.

Unfortunately, only a few of the described cases have been investigated from a molecular point of view. The gene controlling γ-carboxylase is located on chromosome 2p12 and consists of 15 exons that encode a 758 aminoacid peptide.

At least five different mutations in the γ-carboxylase gene have been described (Table 14.24).

These mutations are responsible for decreased glutamate-substrate binding, stabilization, or vitamin K binding. In addition, a few cases due to functional deficiency of the multiprotein complex vitamin K 2,3-epoxide reductase have been reported. Genes on short harm of chromosome 16 have been implicated for the encoding of this enzyme, but the specific genotype–phenotype relation of these mutations has not been fully clarified.

D. Clinical picture

Clinical picture is variable. A first suspicion may arise in infancy, especially in those infants who receive vitamin K prophylaxis and continue to manifest easy bruising.

Both sexes are equally involved. Besides easy bruising, bleeding from umbilical stump, hematomas, hematuria, gastrointestinal bleeding, and

Table 14.24 Mutations and biochemical changes found in the patients who have been studied from a molecular point of view

References	Defect	Exon	Mutation or alteration	Genotype	Comment
Pauli *et al.* (1987)	VKO	/	↑ Level VKO	?	
Brenner *et al.* (1998)	Carbox.	9	Leu394Arg	Homozygous	
Oldenburg *et al.* (2000)	VKO	/	↑ Level VKO	?	Two families
Spronk *et al.* (2000)	Carbox.	11	Try501Ser	Homozygous	
Mousallem *et al.* (2001)	Carbox.	11	Tyr501Ser	Homozygous	
Thomas and Stirling (2003)	Carbox.	1	Deletion (1056–1069)	Homozygous	
Rost *et al.* (2004)	Carbox.	3, 11	Splice site (deletion); Arg485Pro	Comp. heterozygous	
Rost *et al.* (2006)	Carbox.	9, 11	His404Pro; Arg485Pro	Comp. heterozygous	

VKO, Vitamin K epoxide; Carbox., γ-carboxylase; Comp., compound.

menorrhagia has been reported. Brain hemorrhage has also been described. Hemarthroses have never been reported. The clinical picture is similar to that seen in single factor deficiency of the PC. Usually, it is less severe than in severe FII or FX deficiency. The severity of bleeding may vary from time to time without apparent reasons, and patients may have periods with only minor bleeding manifestations. Parents are usually asymptomatic.

There is a limited but variable response to vitamin K administration. Often the administration of vitamin K intravenously or orally has partially or even almost fully corrected the defect decreasing, at least temporarily, the bleeding tendency. Occasionally, patients may appear asymptomatic. Variable skeletal abnormalities, mental retardation, and pseudoxanthoma elasticum have occasionally been seen in the patients.

E. Diagnosis

Laboratory diagnosis may be difficult because of the variability of results during the observation period.

Anti-vitamin K drugs, liver failure, and malabsorption must be excluded. Circulating anticoagulant has also to be excluded. PT-derived tests and PTT

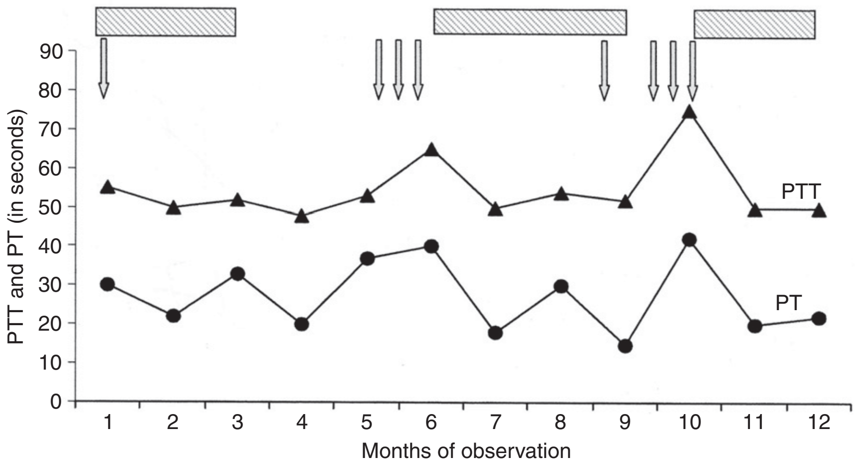

Figure 14.17 **Behavior of PTT (normal 32–40 s) and PT (normal 13–15 s) during a 12-month observation period in a patient with combined deficiency of Factors II, VII, IX, and X. Arrows indicate bleeding episodes. Bars at the top of the figure indicate periods of vitamin K intake (usually oral). Partial and variable correction of clotting pattern is frequently observed in this condition.**

are always, even though variably, prolonged. Thrombin time is normal. Cross-test with equal parts of normal plasmas are normal as expected in the absence of circulating anticoagulants. Protein C and Protein S, when measured, have been found decreased. Parents show a variable pattern that goes from normal to slightly or borderline levels of the four factors. The partial response to vitamin K administration should be of no surprise being typical for most cases with this defect (Fig. 14.17).

The factor levels vary during time, approaching sometimes, in less severe cases, the low normal range. Usually levels vary between 20% and 40% of normal. The levels of the four factors are often decreased to an unequal degree, due to the different half-lives possessed by them. Immunological assays always reveal high levels of antigens as compared with the clotting counterparts.

This is due to the presence of acarboxylated or hypocarboxylated forms in the circulation. Their presence is well demonstrated by a calcium-dependent bidimensional immunoelectrophoresis for FII or FX that shows the presence of two peaks or precipitates, since the acarboxylated forms are more anodic (Fig. 14.18). The same pattern may also be seen by means of a plain immunoelectrophoresis.

Short of molecular biology analysis, the finding of these decarboxylated forms in patients' plasma is the best clue for a correct diagnosis. Needless to say that coumarin drug and vitamin K deficiency from any cause have to be previously excluded. Elevated plasma or serum levels of vitamin K epoxide

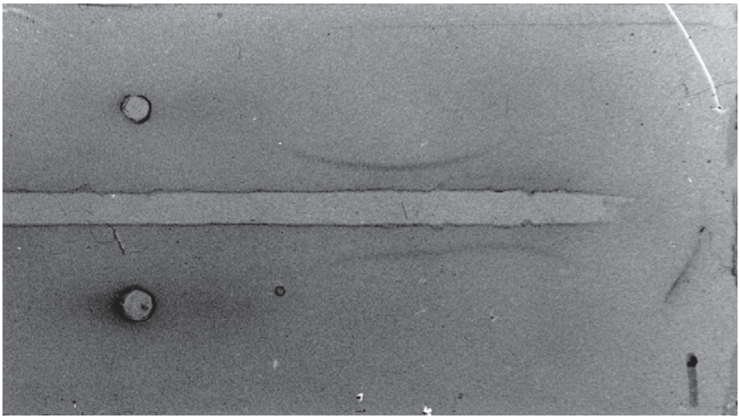

Figure 14.18 Immunoelectrophoresis of plasma of a patient with congenital combined deficiency of Factors II, VII, IX, and X. Upper run normal plasma, lower run patient plasma, anode to the right. In normal plasma only one precipitate is evident whereas two precipitates (normal and acarboxylated prothrombin) are seen in patient plasma.

(vitamin K 2,3) by chemical assays may sometimes be useful (Oldenburg *et al.*, 2000; Pauli *et al.*, 1987).

F. Prognosis and therapy

Prognosis varies with the severity of defect but has to be considered as guarded.

Because of the rarity of the disease, no sure data about survival are available. These patients represent a risk in case of surgical procedures.

Therapeutic measures are limited to continuous vitamin K oral administration. Vitamin K may also be given intravenously. Intramuscolar administration is to be avoided for fear of hematomas. No data are available as to possible differences in response of the oral versus the intravenous administration.

In cases of need, plasma, PC, FVII, or FIX concentrates could be used. Response is good so that surgical procedures can be carried out.

Surgical procedures require levels of at least around 50%. The different half-lives of these factors (about 5 h for FVII and about 72 h for FII) may be responsible for discrepant posttransfusional levels. No data are available about liver transplant or gene therapy.

VII. Deficiency of Protein Z

For the sake of completion, Protein Z deficiency should also be mentioned here, since a few sporadic observations or papers have suggested the possibility that this factor plays a role in bleeding.

Protein Z is a 62,000 single chain glycoprotein vitamin K dependent. It is synthesized in the liver under the coding of a gene located at chromosome 13q34, close to the genes encoding for FVII and FX. It has eight exons and a structure similar to these encoding for the other PC factors (Broze, 2006).

The possibility that Protein Z might be involved in a bleeding diathesis arose from the observations of decreased Protein Z levels in patients with a variable bleeding tendency, but without other clotting defects that the diathesis could be attributed to (Kemkes-Matthes and Matthes, 1995, 2001; Ravi *et al.*, 1998).

Other studies have not confirmed the findings and shown a great variability of Protein Z levels even in the normal population (Gamba *et al.*, 1998). Furthermore, it is interesting to know that no family has been described where both homozygotes (severe deficiency) and heterozygotes (mild deficiency) were present with an established hereditary pattern. Isolated cases with a severe deficiency have also never been reported.

Therefore, the role of protein Z in hemorrhagic disorders seems, at best, not proven and very unlikely.

The limited significance of anecdotical reports, the reported wide range of values in the normal population, the lack of satisfactory family pedigrees, and the vagaries of association studies are sufficient to explain the conflicting results. Therefore, Protein Z is still a clinically function orphan glycoprotein. Needless to say that future studies might clarify to pattern. For example, recent studies on the Protein Z-dependent protease inhibitory might clarify the pattern. In fact, since the activity of this inhibitor seems directed toward FXa, this could supply some support to a bleeding role of this protein at least an indirect one (Broze, 2006).

Protein Z has also been suggested to play both a prothrombotic and/or a protective role in patient with other thrombotic conditions (FV Leiden) on the basis of association studies. However even in this area, results are not univocal and therefore do not seem conclusive (Broze, 2006). This aspect is beyond the scope of this chapter and therefore is not discussed.

VIII. Conclusions

The discussion of the bleeding disorders of the PC is surely important and interesting. However, there is the suspicion that what we know at present is only a part of the picture.

It is remarkable that a vitamin K, widely spread in natural food, might be able to influence so much blood clotting.

Probably, the PC factors are somehow intertwined in their biological functions, the best example of homeostasis. Some factors protect from bleeding, others protect from thrombosis and some, like Protein Z, probably have not found their role yet.

When coumarin drugs are given to a patient, both types of factors are decreased, but, clinically, the protection against thrombosis seems to prevail. However, thrombotic manifestation may occur even in fully anticoagulated patients. Does this mean that, in this setting, Protein C and Protein S deficiency play a role? We do not know.

Does Protein Z play a role? What are the clinically relevant relations between Protein Z, P2-dependent protease inhibitor and FXa?

A role by Proteins C and S is likely, since a quick anticoagulation may cause diffuse ischemic small vessel occlusions with secondary tissue necrosis, namely, "purpura fulminans" in patients severely deficient of Protein C or Protein S.

Is there a regulatory protein responsible for the balance of the two different groups of factors? How do Protein C and Protein S react in the presence of elevated levels of FII or FVII? We just do not know.

The different hereditary pattern shown by this group of factors is fascinating. The fact that only one of them is X-linked, whereas the other five are autosomal remains to be clarified. Does it have a philogenetic significance?

Which of the two first? The sharp difference in biological half-lives shown by these factors, ranging from a few hours (FVII and Protein C) and a few days (FII), is also still lacking a proper explanation. Again, does it have a philogenetic significance?

Altogether, these disorders probably account for only 15–20% of all congenital bleeding disorders (Acharya *et al.*, 2004; Blanchette *et al.*, 1991; Bolton-Maggs and Hill, 1995; Girolami *et al.*, 1985a), hemophilia A, von Willebrand disease, and FXI deficiency accounting for the remaining 80–85%. However, the historical importance of this group of disease is clear.

Their discovery has allowed the construction of an acceptable framework for the understanding of blood coagulation. It is also worthy of mention the fact that the first CRMpos variant ever discovered involved FIX (hemophilia B), followed by FII and FX and, finally, by FVII.

The study of these deficiencies was also instrumental in defining coagulation tests that then became routine, for example, in the follow-up of anticoagulated patients.

Probably the study of patients with these rare coagulation disorders during the past 40 years has also been rewarding at the personal level. On the basis of this experience, we surmise that the contribution of PC factors to blood coagulation is surely not exhausted yet. There is also a recent widespread interest in the extraclotting activity of vitamin K. This followed the observation that carboxylase is present in several tissues. The relation existing between hepatic carboxylase and other tissue carboxylase is still poorly understood. The study of the relations existing between the vitamin K-dependent clotting factors with the other vitamin K-dependent proteins such as osteocalcin and matrix-Gla protein (MGP) may supply further interesting results (Cranenburg *et al.*, 2007). A final comment on the overall results of the molecular biology investigations seems indicated. On the one hand, these techniques have allowed important advances (e.g., the classification of the pathological basis of combined FII, FVII, FIX , and FX deficiency); on the other, they have failed so far to supply definitive answers to the genotype–phenotype relation and/or to improve the therapeutic management of these disorders. The fact that identical mutations may, sometimes, give origin to different phenotypes (e.g., some of those underlying FVII deficiency) or that different mutations in different exons may cause the same phenotype (e.g., those causing hemophilia Bm) remains to be clarified. In addition, it has not been possible so far to indicate with certainty. The mutations, safe for some deletions, associated with CRMpos or CRMneg forms. Refined studies of the purified and/or expressed protein will surely contribute to the understanding and explanation of these discrepancies.

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